SENSACORE®

# ST-200 aQua®

*Electrolyte Analyzer*

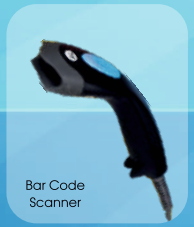
## Bicarbonate




# OPERATOR'S MANUAL

*Analyzer's*
*Advantages*

- *Easy to Operate*
- *Accurate Results*
- *Minimal Maintenance*
- *Economical*


Bar Code Scanner


Mouse

ST-200 aQua®
Electrolyte Analyzer

Bicarbonate

# Operator's Manual

**Sensa Core Medical Instrumentation Pvt. Ltd.**
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The operators should read the entire manual
before installing and operating the analyzer.

# Table of Contents


## 1. Introduction

1.1 Safety....7

1.2 Symbols and Terms....9

1.3 Product Description....11

## 2. Installation

2.1 Installation Requirement....17

2.2 Power Cable Connection....18

2.3 Barcode Scanner Connection....19

2.4 External Grounding Cable Connection....19

2.5 Printer Paper Installation....20

2.6 Reagent Pack Installation....22

2.7 Reference & Electrode Installation....23

2.8 Power ON....25

2.9 Initiating Start-up Process....27

## 3. Instrument Settings

3.1 Screen Menu....30

3.2 Buttons & Icons....33

3.3 Keyboards....39

3.4 Multi Language....40

3.5 Menu Access....41

## 4. Sample Analysis

4.1 Patient Samples....83

4.2 QC Samples....89

## 5. Data base

5.1 Calibration Data....93

5.2 Patient (Sample/Urine) Data....95

5.3 QC Data....98

5.4 Parameters....100

5.5 L.J (Levey Jennings) Graph....101

**6. Standby & Shutdown**

6.1 Standby mode..... 104

6.2 Aalyzer Shutdown..... 105

**7. Maintenance**

7.1 Maintenance Information.....108
7.2 Cleaning.....108

**8. Troubleshooting**

8.1 Troubleshooting.....111

8.2 Display Messages..... 120

**9. Specifications**

9.1 Operating Specifications..... 125

9.2 Instrument Specifications..... 126

9.3 Reagent pack Specifications.....127

9.4 Performance Specifications..... 128

9.5 Principles of Operation..... 129

9.6 Calculated Parameters.....130

**Appendices**

Warranty..... 133

# 1. Introduction

1.1 Safety.....7
1.2 Symbols and Terms.....9
1.3 Product Description.....11

# 1. Introduction

This manual provides all necessary instructions for the routine operation of the **ST-200 aQua Bicarbonate** Electrolyte Analyzer. Please read this manual carefully. It has been prepared to help you attain optimum performance from your Analyzer.

***WARNING**: Blood samples and blood products are potential sources of infectious agents. Handle all blood products and flow path components with care. Gloves and protective clothing are recommended when performing maintenance and troubleshooting procedures.*

***About this Manual:***
*This manual is for the ST-200 aQua Bicarbonate Electrolyte Analyzer.*
***Throughout this manual, NOTE:** indicates especially important information.* ***CAUTION:** indicates information that is critical to avoid instrument damage or incorrect results*, and ***WARNING:** indicates a possible hazard to the operator.*

## 1.1 Safety

Personnel operating this analyzer must be proficient in the operating and replacement procedures of the analyzer. The following safety procedures must be followed.

**General Safety**

- Read the safety and operating instructions before operating the analyzer.
- Follow all operating and use instructions.
- Use only the provided power cord and adapter.
- Place the analyzer so that its location or position does not interfere with its proper ventilation.
- Place the analyzer away from heat sources.
- Connect the analyzer to a power supply only of the type described in the operating instructions or marked on the analyzer.
- Do not use a damaged power cord, adapter, or loose outlet. Never touch the power supply accessories with wet hands. Keep dust away from the outlet and power cord.
- Do not use or store any hot equipment or flammable material near the analyzer or power supply accessories.
- Do not allow water or foreign substances to get into the analyzer or power supply accessories.
- Do not directly look at the red light emitting from the bar-code scanner as direct eye exposure to this light can damage vision.

**Safety, *continued***

**General Caution**

- The Power cord must be plugged into a grounded outlet.
- The rear vent of the analyzer must be free of obstruction and not covered by cloth or any other material.
- Do not install or operate the analyzer in an area where ferro magnetic fields are generated.
- Before long term storage of the analyzer, remove the Reagent Pack, turn off the power, and clean any contamination from the analyzer.
- There are no operator serviceable parts inside the product. Do not disassemble, repair, or modify the product.
- If Electro-mechanical problems are suspected, call a service engineer to report the problems.
- When disposing of the analyzer and provided electrical accessories, contact your local distributor where you purchased them.
- You must not discard this electrical/electronic product in domestic household waste.

**Bio Hazard Caution**

- All materials used in collecting blood and/or other samples from humans should be treated as bio-hazardous materials with the potential to carry infectious agents capable of producing disease.
- All bio-hazardous materials should be handled and disposed of in accordance with applicable rules and regulations of the hospital, laboratory, or other testing facilities.
- Operate the analyzer in the appropriate environment.
- Wear appropriate laboratory attire, e.g., gloves, lab coat, and breathing apparatus, when working with hazardous materials.
- Be careful not to let the sample and/or collecting tools directly touch the mouth, eyes, mucus membranes, or any area with broken skin.
- Wash hands after collecting the sample or using the analyzer.

## 1.2 Symbols and Terms

### 1.2.1 Symbols

The symbols described below are used on the packaging of ST-200 aQua Bicarbonate Electrolyte Analyzer related products.

| Symbol | Description |
|---|---|
|  | This product fulfils the requirements of Directive 98/79/EC on in-vitro diagnostic medical devices. |
|  | ***In Vitro*** diagnostics medical device |
|  | Please read pack insert / Follow the instrument's instructions for use! |
|  | Serial number |
|  | Caution: Attention, see operating instructions |
|  | Manufactured by |
|  | Temperature Limit Symbol – Products and/or components which contain this symbol must be stored within the specified temperature range. |
|  | Catalog number |
|  | Batch Code Symbol – Manufacturing lot number is located to the right of this symbol. This symbol is located on all products, which are controlled via a lot number. |
|  | Expiration / Use By Symbol – Product to be used by the expiration date indicated to the right of this symbol. This symbol is located on all consumables, which are controlled via an expiration or use by date. |

### 1.2.2 Terms

| Term | Description |
|---|---|
| **$Na^+$** | Sodium ion |
| **$K^+$** | Potassium ion |
| **$iCa^{++}$** | Ionized Calcium |
| **$Li^+$** | Lithium ion |
| **pH** | Potential of Hydrogen |
| **$Cl^-$** | Chloride ion |
| **$HCO_3^-$** | Bicarbonate Level |
| **$pCO_2$** | Partial pressure of Carbon dioxide |
| **$TCO_2$** | Total Carbon Dioxide |
| **BE** | Base Excess |
| **nCa** | Normalized Calcium |
| **TCa** | Total Calcium |
| **Ag-Na** | Anion gap of Sodium |
| **Ag-K** | Anion gap of Potassium |

## 1.3 Product Description

### 1.3.1 Intended Use

The ST-200 aQua Bicarbonate Electrolyte Analyzer is an automated microprocessor-controlled analyzer for measurement of Sodium, Potassium, Ionized Calcium, pH, Chloride, Bicarbonate & Lithium in Whole Blood, Electrolytes in Serum/Plasma/CSF, and Diluted Urine. Measurements obtained by this device are used for the diagnosis, monitoring, and treatment of diseases involving electrolyte imbalance.

To obtain accurate results, the ST-200 aQua Bicarbonate must be operated with Our Company's specially packaged Calibrant and proven-based control materials and be maintained as described in this manual.

### 1.3.2 Measured Parameters

**ST-200 aQua Bicarbonate Electrolyte Analyzer:** $Na^+$, $K^+$, $iCa^{++}$, $Cl^-$, pH, $HCO^-_3$ and Lithium (Optional)

**$Na^+$** Sodium measurement is used in the diagnosis and treatment of aldosteronism, diabetes insipid us, adrenal hypertension, Addison's disease, dehydration, or diseases involving electrolyte imbalance.

**$K^+$** Potassium Measurement is used to monitor electrolyte balance in the diagnosis and treatment of disease conditions characterized by low or high potassium levels.

**$Cl^-$** Chloride measurement is used in the diagnosis and treatment of electrolyte and metabolic disorders such as cystic fibrosis and diabetic acidosis.

**$iCa^{++}$** Calcium measurements are used in the diagnosis and treatment of parathyroid disease, a variety of bone diseases, chronic renal disease and tetany (intermittent muscular contractions or spasms).

**$HCO^-_3$** Bi-Carbonate measurement is used in the diagnosis and treatment for pH levels in your blood and for kidney disease, liver failure, or other conditions related to metabolism.

**pH** pH is a measure of hydrogen ion concentration are used in the diagnosis and treatment for Kidney disease, Lung disease and Asthma.

**$Li^+$** Lithium measurement are used in the diagnosis and treatment for muscular weakness, Kidney and for mental disorder.

### 1.3.3 Calculated Parameters

pCO2, $TCO_2$, nCa, TCa, BE, AG_Na, AG_K

### 1.3.4 The Sample

Lithium heparin or Sodium heparin whole blood samples from syringes, open tubes, and capillary tubes can be used on the ST-200 aQua Bicarbonate Electrolyte Analyzer. The minimum sample size for analysis is 120 µL.

### 1.3.5 Suitable Anticoagulants

- Lithium heparin and Sodium heparin are the acceptable anticoagulants for use with the analyzer.
- EDTA, citrate, oxalate, and sodium fluoride **ARE NOT** acceptable for use.
- Depending on the amount of heparin used in the collection syringe and whether it is filled to capacity with blood, heparin concentrations of the result may vary.
- Liquid or dry heparin when present in **excess** may **cause errors.** Ensure blood collection devices are filled as per manufacturer instructions.

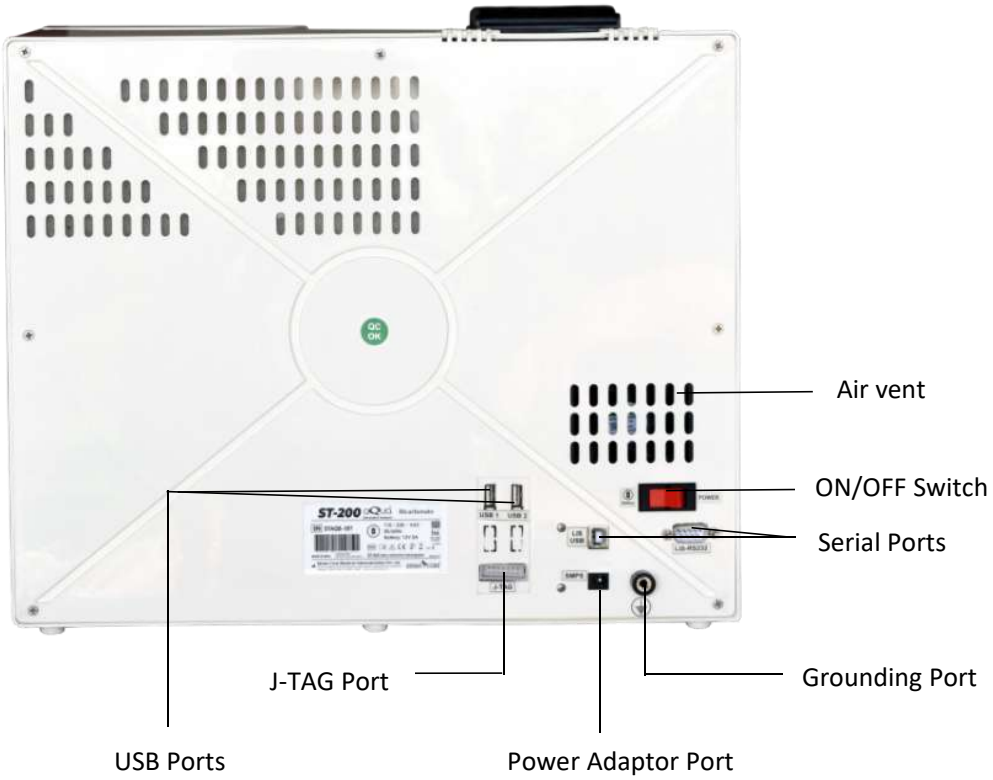
***CAUTION:***

*ST-200 aQua Bicarbonate Electrolyte Analyzer users should take careful note of these considerations when establishing reference intervals and interpreting results.*

### 1.3.6 Analyzer




- **Touch Screen** guides users through all aspects of analyzer operation, including analyzer setup, information entry and display, and testing.
- The screen is powered on by pressing the power button of the touch screen panel.
- **Printer** provides hard copies of test results and information.
- **Sample Cover/Flip** aspirates the sample for analysis.
- **Reagent Pack** is used for calibration and analysis of parameters.


Air vent
ON/OFF Switch
Serial Ports
J-TAG Port
Grounding Port
USB Ports
Power Adaptor Port


Sample Tube
Bubble Detector
Flip
Reference Housing
Reference Electrode
External Waste Tube
Pump Tubing
Pump Module
Pinch Valve Module
V-1
V-3
V-2

### 1.3.7
### Reagent Pack




### 1.3.8
### Accessories



Power Adapter



Power Cord



Paper Rolls

# 2. Installation

2.1 Installation Requirement.....17
2.2 Power Cable Connection..... 18
2.3 Barcode Scanner Connection..... 19
2.4 External Grounding cable Connection.....19
2.5 Printer Paper Installation ..... 20
2.6 Reagent Pack Installation..... 22
2.7 Reference & Electrode Installation..... 23
2.8 Power-On.....25
2.9 Initiating Start- up Process..... 27

## 2.1 Installation Requirement

Before you begin installing your ST-200 aQua Bicarbonate Electrolyte Analyzer, take a moment to look over the contents to ensure that you have the following:

- Power supply with power cord
- Quality Control materials
- ST-200 aQua Bicarbonate Reagent Pack
- Thermal printer paper rolls
- Reference Electrode Kit
- Troubleshooting Kit

### 2.1.1 Unpacking

- Inspect the shipping package of the product for any signs of damage from shipment.
- Ensure all listed components are included in the package.

### 2.1.2 Environmental Requirements

**Working Area Requirements (Environmental):**

Keep the working area around the system free of dirt, corrosive fumes, vibration, and excessive temperature changes.

- Location : Indoors
- Operating Temperature : 15 - 35 °C
- Operating at Humidity : < 85 %

***Note:***

- ✓ The analyzer should be installed on a horizontal, stable surface in a well-ventilated area without direct sunlight or heat.
- ✓ Allow enough space to place the reagent pack in the reagent pack slot.
- ✓ Allow at least 10 cm of space from the back of the analyzer for adequate ventilation.

### 2.1.3 Electrical Requirements

- Operating Voltage Range : 220 V AC
- Amp : 0.75 A
- Operating Frequency : 50-60 Hz
- Power cord : 3 wire
- Power Adapter : Output- 12V, 5.0A, 60 W MAX

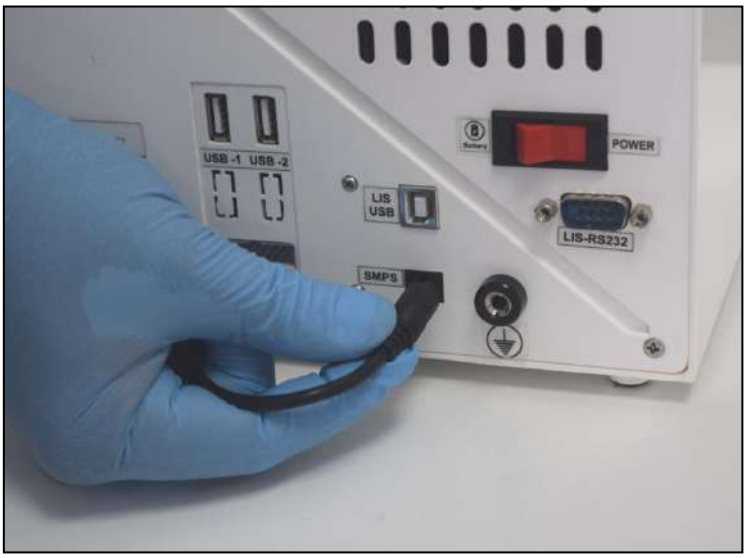
## 2.2 Power Cable Connection

***Caution***

- *Check that the power switch is in OFF position.*
- *Always connect the power according to the steps described in this manual.*
- *Always use a grounded electrical outlet.*
- *Use only the power cord and power adapter provided by the manufacture.*
- *Electrical plugs and sockets vary by countries. If necessary, use appropriate power plug or power strips to connect the provided power cord to the outlet.*

**Plug in the Power Supply**

1. Plug the power cord into the socket of the power adapter.
2. Plug the power supply into the receptacle on the back side of the analyzer.




3. Plug the power cord into an grounded electrical outlet.

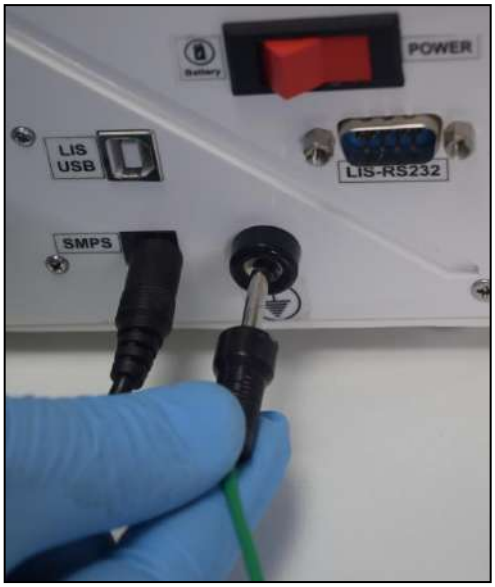
## 2.3 Bar-code Scanner Connection

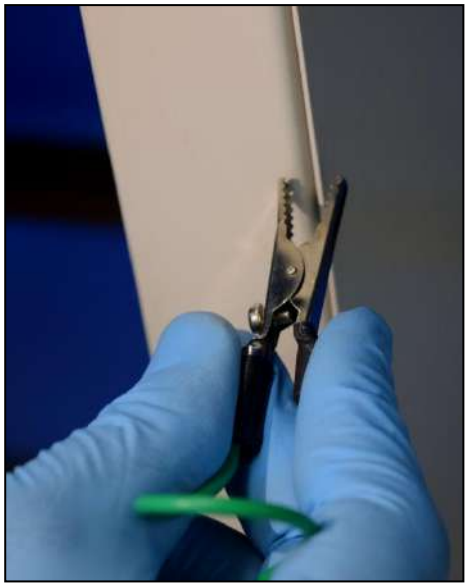
1. Insert the USB connector of bar-code scanner into a USB port on the backside of the analyzer.




## 2.4 External Grounding Cable Connection

1. Connect the External Grounding cable into a Grounding port on the back side of the analyzer.
2. Connect other end of the cable to any of the metal part.



## 2.5 Printer Paper Installation

**Install Printer Paper**

1. Facing the top of the analyzer, pull the lever of the printer to open the cover.



2. Place the paper roll in paper compartment.

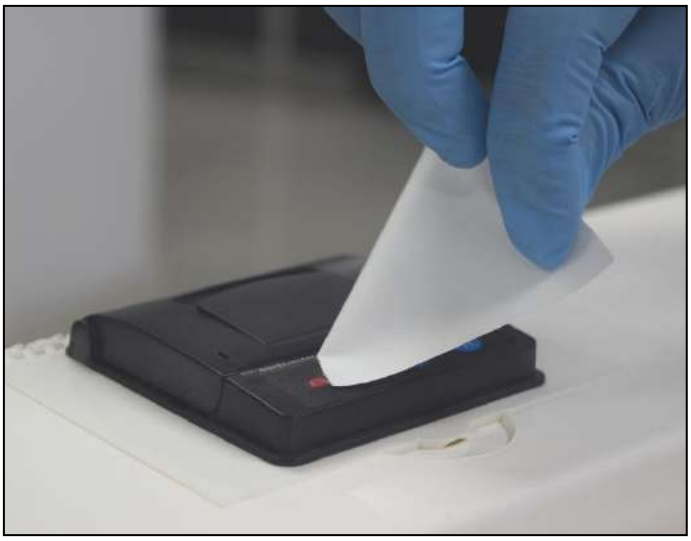


3. Fix the paper roll in the paper compartment such that end of the paper is outside the printer.

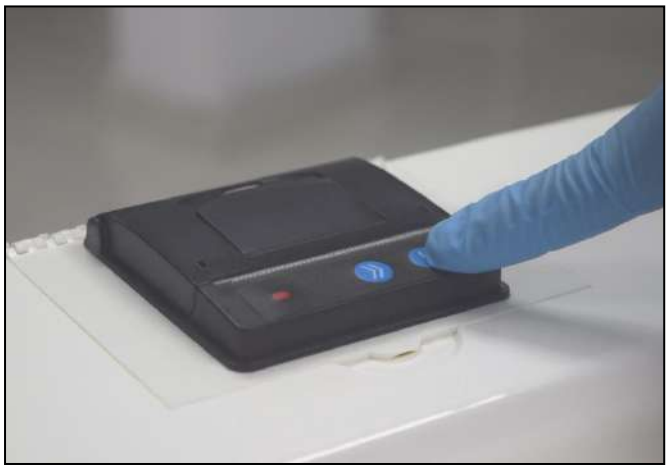
## Printer Paper Installation, *Continued*

**Remove Printer Paper**

1. Flip down the printer cover (Press hard) and remove the remaining Paper.



2. Hold down the pause button to advance the paper.



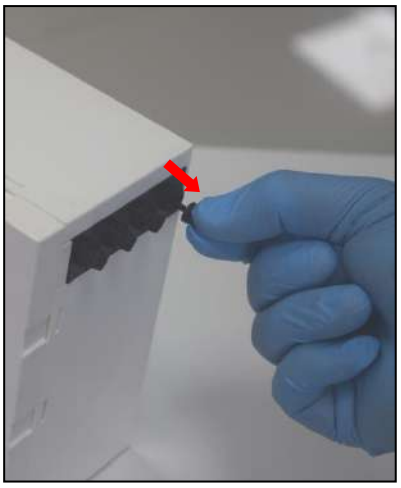
***Note:***

- ✓ Do not use force to pull out the paper from the printer.
- ✓ Never pull the paper in the reverse direction of printing.
- ✓ The Feed button is only active when the printer detects paper in the printer.

## 2.6 Reagent Pack Installation

**Install Reagent Pack**

1. Unpack Reagent pack from the packing box.
2. Take out the stoppers from reagent pack.



3. Facing the reagent pack label outside, insert the reagent pack into the Analyzer through guiding slot.



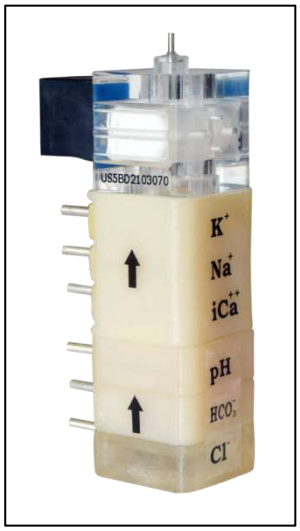
**Note:**

Do not place the ST-200 aQua Bicarbonate reagent pack open to atmosphere, ensure to insert stoppers when reagent pack left ideally.

## 2.7 Reference & Electrode Installation

**Electrode Installation**

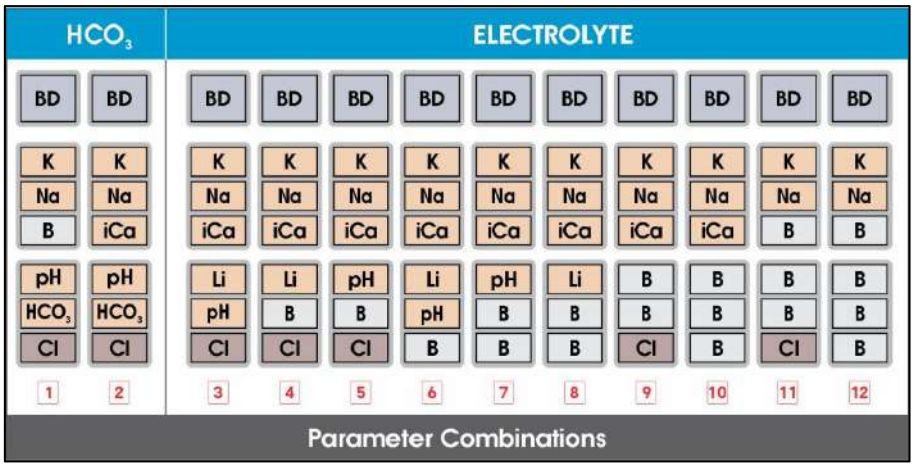
1. Unpack electrodes, seals and reference housing from packing box.
2. Stack the electrodes according to Electrode parameters by inserting seals in between every Electrode, also connect sample detector.




3. Slide the staged electrode set in Reference Housing. Push Down firmly until the sample detector snaps into its position and insert the Reference lock to complete the staging.
4. Connect the sample tube to the top of the bubble detector.
5. Connect the pump tube to the base of the electrode housing.

**Note:**

**Electrode Parameters**

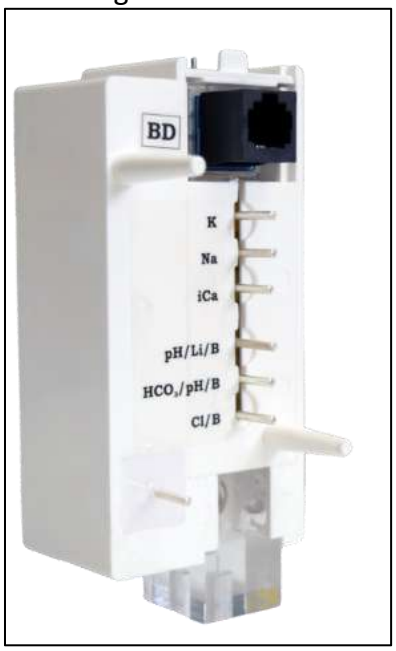



**Removal of electrodes**

✓ Slide the electrode stack up at least one inch by gently pushing up on the bottom electrode pin. This process will remove electrodes from the reference housing.




**FRONT VIEW**




**BACK VIEW**

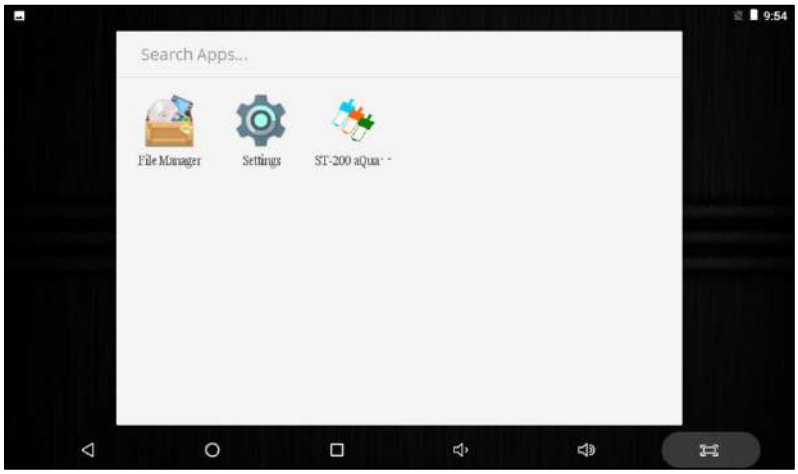
## 2.8 Power ON

***Caution***

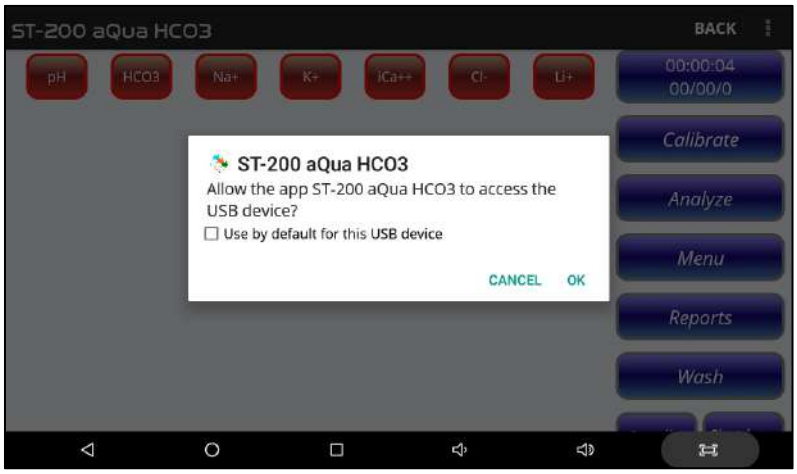
- *Remove any attached USB memory device from the analyzer before turning the power switch ON.*

### 2.8.1 Turn on the power

1. Facing the back of the analyzer, flip the Red power switch on the lower left side, to the "**ON**" position to start the machine. Give a long Press to the Tab Power switch to the 'ON' touch screen.
2. The screen will turn on and the analyzer starts booting. The **"Booting… Please Wait"** message appears on the screen.
3. After booting is finished, open applications to start ST-200 aQua $HCO_3$ application.

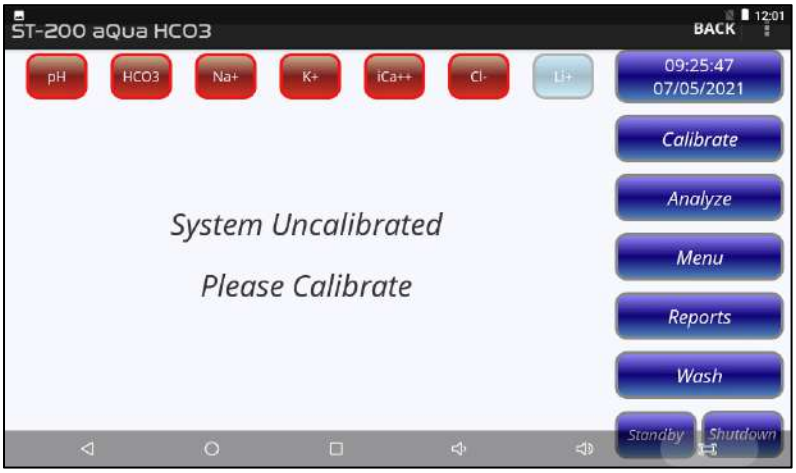



4. Select **ST-200 aQua $HCO_3$ App**, it asks for the acceptance of USB device access.



**Turn on the Power,** ***Continued***

5. Press OK, **"ST-200 aQua $HCO_3$ Analyzer Ver: xxxx"** message displays on the screen.
6. System starts initializing by displaying **"Initializing"** message on the screen.
7. After initializing it automatically performs Self Check. The **"Self Check"** message displays on the screen.
8. System asks to calibrate, by displaying the following screen.




9. During the first time of installation, the user is advised to perform a new reagent pack option in the menu. It performs priming followed by calibration. If any of the parameters is uncalibrated, re-calibrate the instrument. Calibrate the analyzer by pressing Calibrate icon on the main screen.

***Note:***

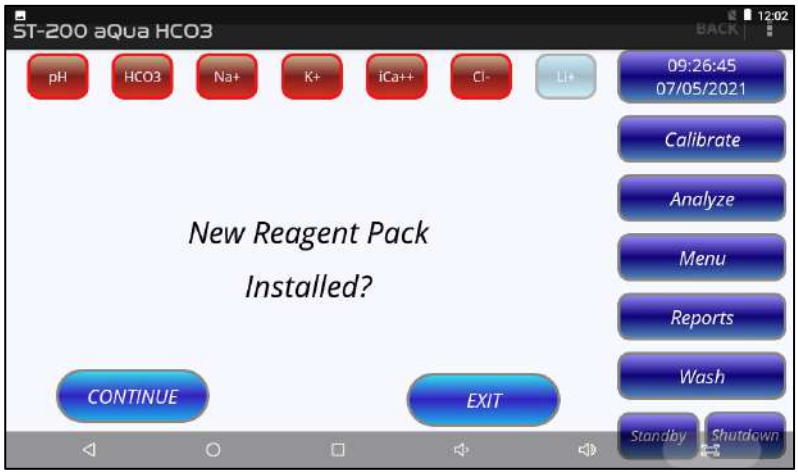
✓ Calibration is required for Sample Analysis.

## 2.9 Initiating the Start-up Process

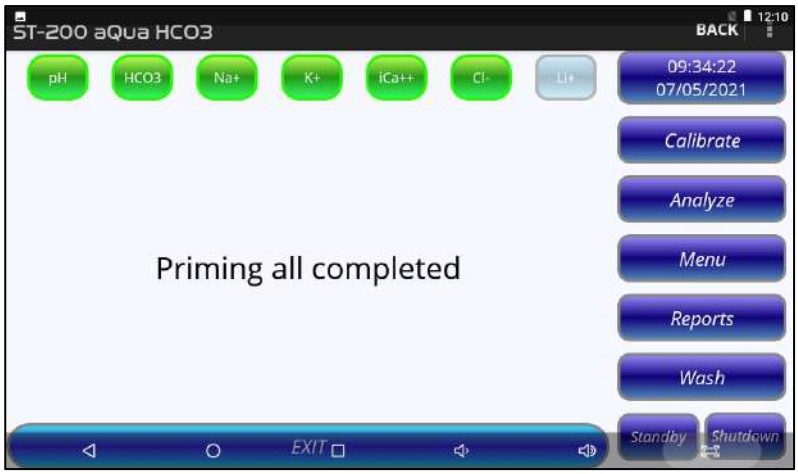
After completing the whole flow path connections from the reagent pack to the waste outlet via reference housing. Perform, New Reagent pack option which does reagent priming and calibration.

### 2.9.1 Start-up process

1. Go to **Menu** and select **New Reagent Pack**. Following screen will appear.




2. Press **Continue.** During the process, following messages gets prompted on the screen ,***"Priming Cal B Please wait", "Priming Cal C Please wait" & "Priming Cal A Please wait"***. After priming the following screen will appear.



**Start-up process *Continued,***

- ✓ After reagent pack is primed system will automatically calibrate.

**Note:**

- ✓ The other way to prime the reagents is as follows
  - Go to **Menu > Diagnostics > Flow > Reagent priming**
  - Select Prime Cal A for priming Cal A reagent.
  - Select Prime Cal B for priming Cal B reagent.
  - Select Prime Cal C for priming Cal C reagent.
  - Select Prime All for priming all reagents.

Press **Exit** to exit the present screen.

# 3. Instrument Settings

3.1 Screen Menus.....30
3.1.1 Ready Screen.....30
3.1.2 Menu options.....31
3.1.3 Wash.....32
3.2 Buttons and Icons.....33
3.3 Keyboards.....39
3.4 Multi language.....40
3.5 Menu Access.....41
I. Diagnostics.....43
1. Daily Cleaner.....43
2. Maintenance.....45
3. Sensors.....46
4. Flow.....48
5. Hardware.....58
6. Reagent Pack.....61
II. User Menu.....64
2.1 Normal Ranges.....65
2.2 Calibration Frequency.....69
2.3 Sipping Frequency.....69
2.4 Auto Standby.....69
2.5 Correction Factor.....70
2.6 Precision.....72
III. Setup Menu.....74
3.1 Analyze Type.....75
3.2 Date and Time.....78
3.3 Printer.....79
3.4 Print-mV.....79
3.5 Patient-ID.....80
3.5 Print Parameters.....81

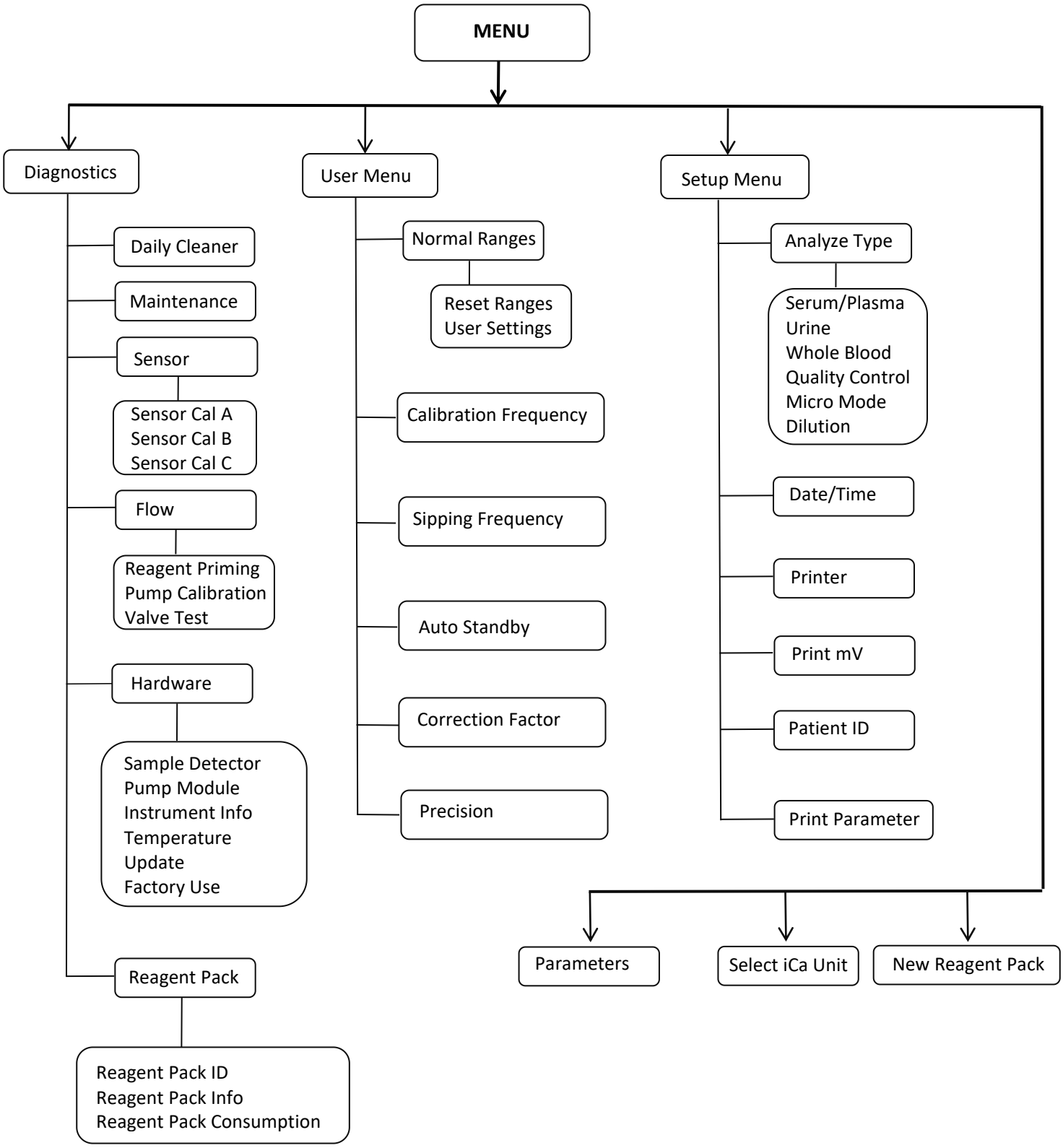
## 3.1 Screen Menus

### 3.1.1 Ready Screen




| No. | Description |
|---|---|
| 1 | Parameters (Calibration status of electrodes) |
| 2 | To Change language, background image and android software version. |
| 3 | Current date and time of the analyzer |
| 4 | To perform Calibration |
| 5 | To Analyze the sample |
| 6 | Menu access |
| 7 | Reports of performed Analysis and calibration results |
| 8 | Wash (described below) |
| 9 | To Shutdown |
| 10 | Stand by Mode |
| 11 | To return to the home/idle screen or exit the present process. |

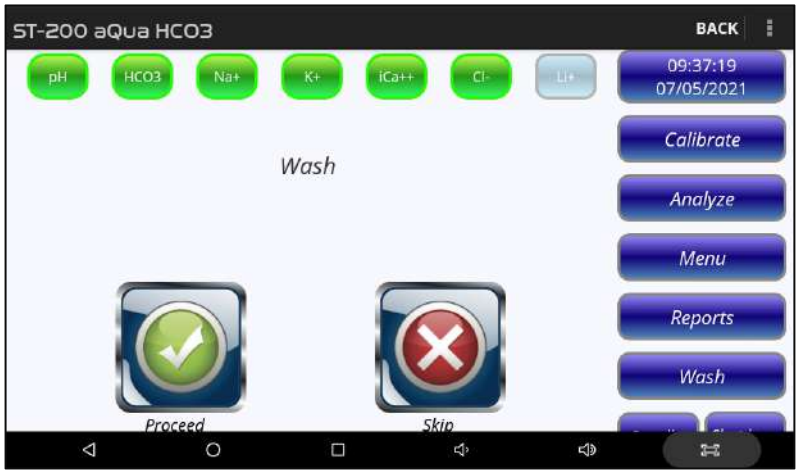
### 3.1.2 Menu options



### 3.1.3 Wash

- Perform Daily Cleaner solution in Wash. Following is the procedure of Wash.

1. Select **Wash** from the home screen, following screen is displayed.




2. Press Proceed then instrument shows **"Please Lift Sampler"** message.
3. Lift the sampler and place cleaner solution into the Probe, press **Continue** for aspirating the solution.
4. **"Aspirating Pl*ease wait"*** message appears on the screen while aspirating the sample. Instrument produces a beep sound after aspirates sufficient solution.
5. Instrument prompts to remove cleaner sample and close the sampler. Then, it performs washing process by displaying ***"Performing washing Please wait"*** followed by calibration.

## 3.2 Buttons and Icons

| | |
|---|---|
|  Diagnostics | Go to Diagnostics Menu |
|  Daily Cleaner | Select to run Daily cleaner |
|  Maintenance | Go to Maintenance |
|  Sensor | Go to sensor Menu |
|  Sensor Cal A | To check Sensor Cal A Values |
|  Sensor Cal B | To check Sensor Cal B Values |
|  Sensor Cal C | To check Sensor Cal C Values |
|  Flow | Go to Flow Menu |
|  Reagent Priming | To prime Reagent |
|  Priming Cal A | To prime Cal A Reagent |
|  Priming Cal B | To prime Cal B Reagent |
|  Priming Cal C | To prime Cal C Reagent |

**Buttons and Icons *Continued,***

| | |
|---|---|
|  **Prime All** | To prime all Reagents |
|  **Pump Calibration** | To check Pump Calibration |
|  **Valve Test** | Go to Valve test |
|  **Cal A Flow** | To check Cal A valve Flow |
|  **Cal B Flow** | To Check Cal B valve Flow |
|  **Cal C Flow** | To Check Cal C valve Flow |
|  **Air Flow** | To Check Air valve Flow |
|  **A/B Flow** | To Check Cal A/B valve Flow |
|  **Cal A/B/Air Flow** | To Check Cal A/B/Air valve Flow |
|  **Hardware** | Go to Hardware menu |
|  **Sample Detector** | Go to Sample detector mode |
|  **Pump module** | To check Pump Module |

**Buttons and Icons** ***Continued,***

| | |
|---|---|
|  Instrument Info/Software Version | Go to software version screen |
|  Temperature | To check Temperature |
|  Update | To update APK file |
|  Factory Use | For Factory Use |
|  Reagent Pack | Go to Reagent Pack Menu |
|  Reagent Pack ID | To check Reagent pack ID |
|  Reagent Pack info | To know Reagent Pack Information |
|  Reagent Consumption | To check the reagent pack consumption |
|  User Menu | Go to User menu |
|  Normal Ranges | Go to Normal Ranges Menu |
|  Reset Ranges | To Reset Ranges |
|  User Settings | Go to User Settings |

**Buttons and Icons** ***Continued,***

| | |
|---|---|
|  Sample Ranges | To edit sample ranges |
|  Urine Ranges | To edit urine Ranges |
|  QC 1 Range | To select Quality control Level 1 |
|  QC 2 Range | To select Quality control Level 2 |
|  QC 3 Range | To select Quality control Level 3 |
|  Calibration Frequency | For setting calibration frequency |
|  Sipping Frequency | Select to set sipping frequency |
|  Auto Standby | Go to Standby Mode |
|  Correction Factor | For setting correction factor of Parameter |
|  Precision | For calculating Electrodes precision |
|  Setup Menu | Go to setup Menu |
|  Analyze Type | To select the Analyze sample |
|  Serum-Plasma | Go to Serum-Plasma Mode |

**Buttons and Icons *Continued,***

| | |
|---|---|
|  Urine | To select urine mode |
|  Whole Blood | Go to Whole Blood Mode |
|  Quality Control | Go to Quality Control Mode |
|  Micro Mode | Go to Micro mode |
|  Dilution | Go to Dilution mode |
|  Date & Time | Select to set Date and Time |
|  Printer | To select the print status |
|  Print mV | To select the print unit |
|  Patient ID | To select Patient ID ON/OFF status |
|  Print Parameter | To print the required Parameter |
|  Parameters | Go to Parameters Menu |
|   Select iCa Unit | To select iCa units |
|  New Reagent pack | To install new reagent pack |

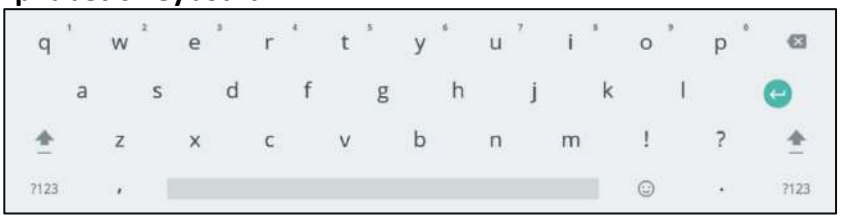
**Buttons and Icons** ***Continued,***

| | |
|---|---|
|  **Clear Count** | For use of organization |
|  **OK** | To continue |
|  **Exit** | To Exit |
|  **Shutdown** | To shut down the system |
|  **Skip** | To skip shutdown |

## 3.3 Keyboards

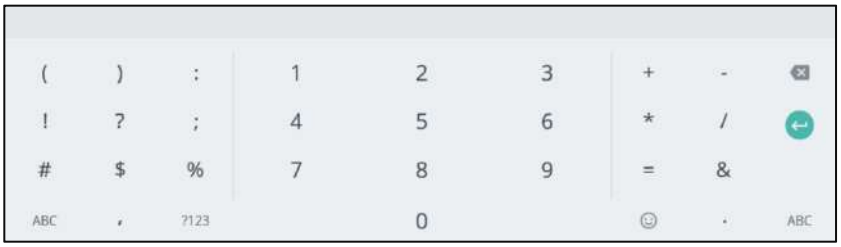
- The analyzer offers following keyboards for data entry.

**Alphabetic Keyboard**




**Numeric & Special Character Keyboard**



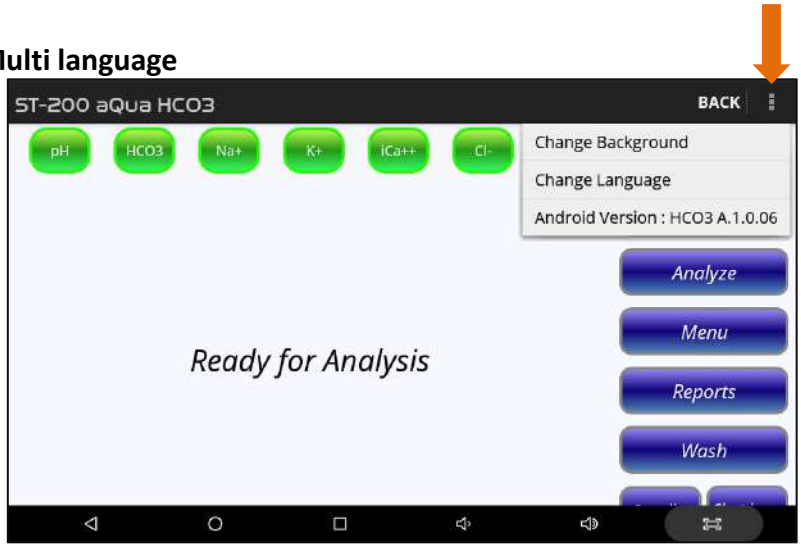

**Date and Time Entry Keyboards**



## 3.4 Multi language

- The analyzer offers 91 languages. User can select their respective languages from the list. For language selection, Press the three dots beside back button. A drop down menu appears, select Change Language.

**1. Multi language**




**2. Select your language from the scroll list.**




## Change Background

- Allows user to change the background image of idle screen/Home screen.

## Android Software Version

- Allows user to know the updated android software version.

## 3.5 Menu Access

**Menu :** Select **Menu** on the main menu screen. The Menu options will appear.




1. **Diagnostics:** For the Diagnostics menu options, press **Diagnostics**.

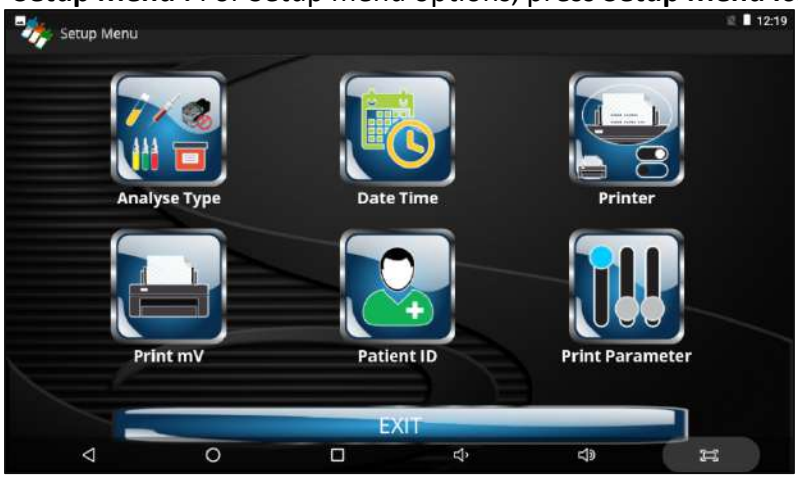



2. **User Menu :** For user menu options, press **User Menu.**

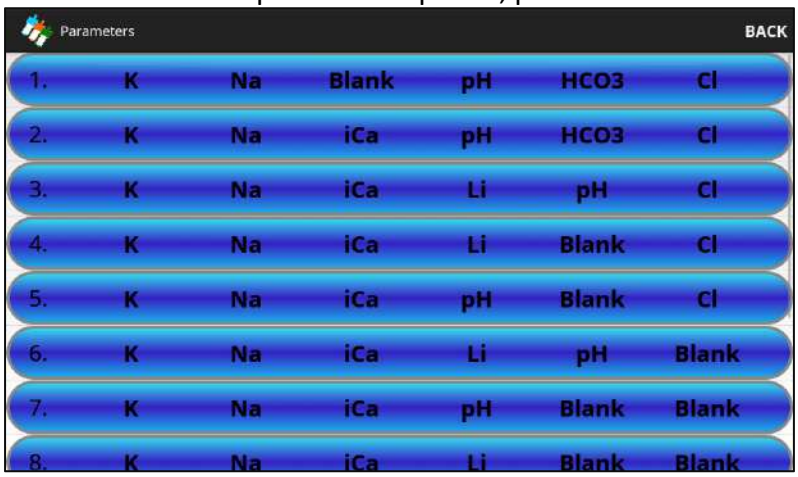


**Menu Access** ***Continued,***

3. **Setup Menu :** For Setup menu options, press **Setup Menu Icon.**




4. **Parameters :** For parameter options, press **Parameters Icon.**




***Note:*** *Select the required parameter.*

5. **Select iCa unit:** For changing the unit of iCa (mg/dL or mmol/L).




6. **New Reagent Pack:** To perform self priming followed by calibration, when new reagent pack is inserted/ installed.

# I. Diagnostics

## 1. Daily Cleaner

The fluid path must be cleaned after every 25 samples to remove protein deposition from the sensors. The daily cleaner is required and is essential for trouble-free analyzer operation.

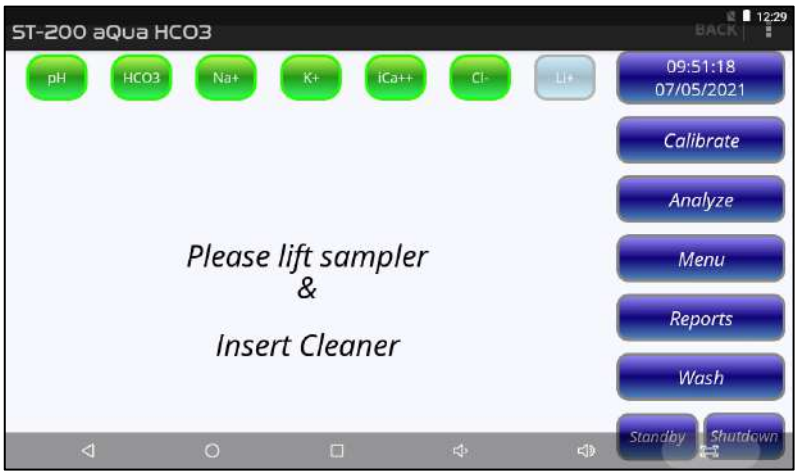
- Following is the procedure of daily cleaner.

**Daily Cleaner**

1. For the Daily Cleaner options,
   Go to **Menu > Diagnostics > Daily Cleaner**.



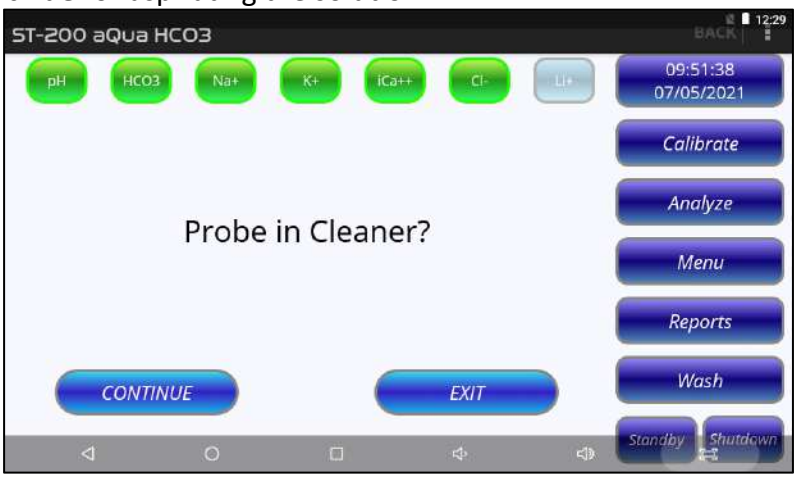

2. Instrument displays the following Screen.



## Diagnostics *Continued,*

**Daily Cleaner, *Continued***

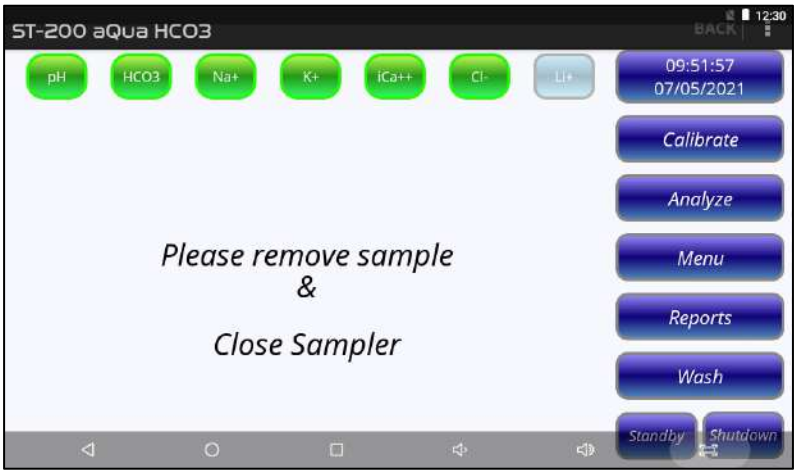
3. Lift the sampler and place cleaner solution into the Probe, press **Continue** for aspirating the solution.




4. ***"Aspirating Please wait"*** message appears on the screen while aspirating the sample. Instrument produces a beep sound after aspirates sufficient solution.




3. Instrument prompts to remove cleaner sample and close the sampler. Then, it performs cleaning process by displaying ***"Cleaning performing Please wait"*** followed by calibration.



**Diagnostics *Continued,***

## 2. Maintenance

Whenever operator wants to clean the instrument flow/Change electrodes/Change tubing's or any disassembling of flow modules. Then, operator should activate the maintenance mode which clears the liquid present in the tubing's.

**Maintenance**

1. Go to **Menu** > **Diagnostics**
2. Select **Maintenance**. The following screen will appear.




3. The **"Maintenance"** message will appear on the screen after few seconds. The maintenance mode is activated.




4. To exit maintenance press **Exit**.
5. The ***"Exiting Maintenance"*** message appears on the screen.
6. Press **Exit** to exit to the main menu screen.

**Note:**

- ✓ After exiting maintenance system asks to calibrate by displaying the message ***"System Uncalibrated Please Calibrate"***

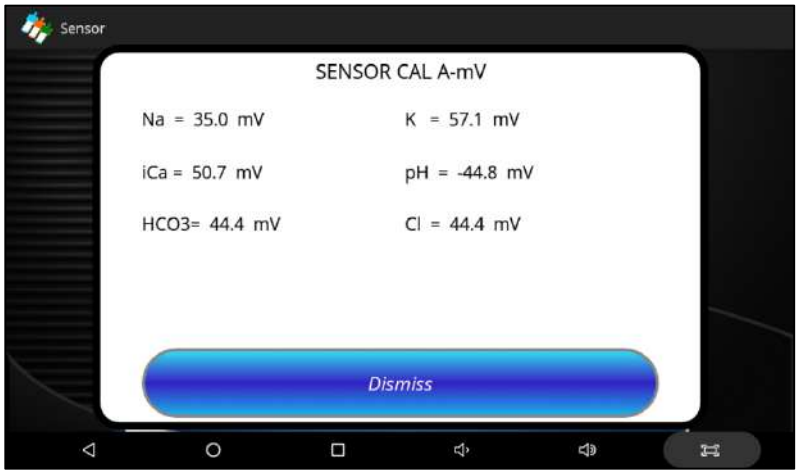
**Diagnostics *Continued,***

## 3. Sensors

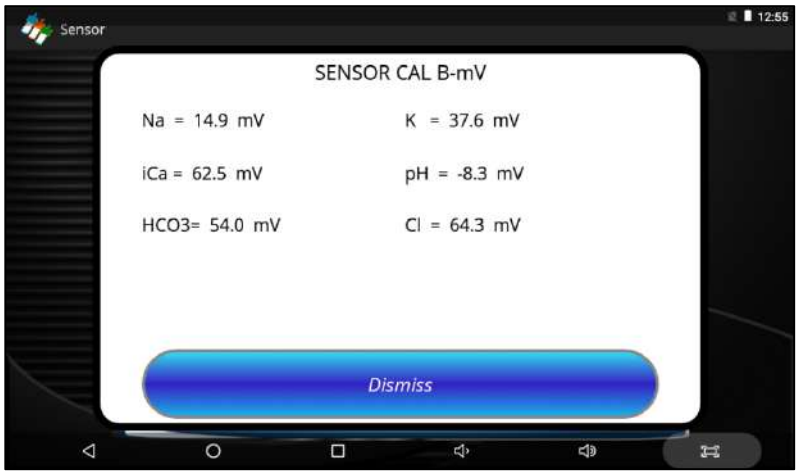
Sensors is used to check mV of the electrodes for each reagent (CAL A /CAL B/CAL C) separately.

**Sensors**

1. Go to **Menu > Diagnostics > Sensor**
2. Select **Sensor Cal A**. The ***"CAL A Measurement"*** message will appear on screen. After few seconds following screen appears.




3. Press **Dismiss** to exit screen to main menu screen.
4. Select **Sensor Cal B**. The ***"CAL B Measurement"*** message will appear. After few seconds following screen will appear.

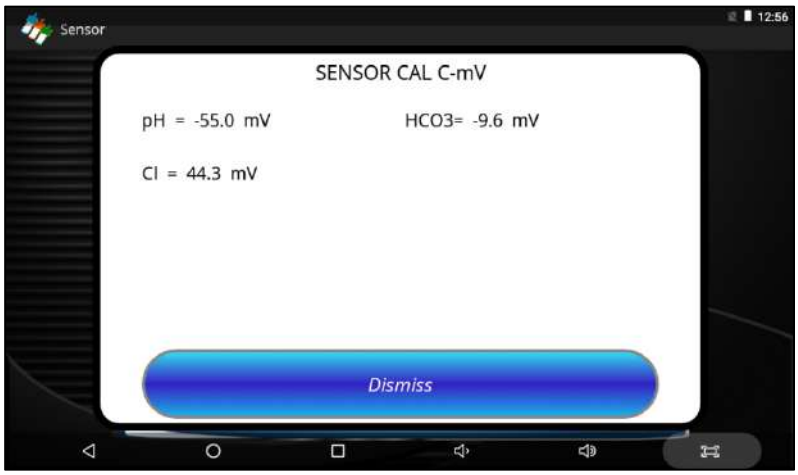



5. Press **Dismiss** to exit screen to main menu screen.

**Diagnostics *Continued,***

**Sensors, *Continued,***

6. Select **Sensor Cal C**. The ***"CAL C Measurement"*** message will appear. After few seconds the screen displays as follows




7. Press **Dismiss** to exit screen to main menu screen.

**Diagnostics *Continued,***

## 4. Flow

It is used for initializing or troubleshooting flow issues in the instrument.

**4.1 Reagent Priming :** It is used to prime the liquid from reagent pack.

4.1.1 Prime Cal A
4.1.2 Prime Cal B
4.1.3 Prime Cal C
4.1.4 Prime All

**4.2 Pump Calibration :** It is used to know the pump tubing durability.

**4.3 Valve Test :** It is used to check the valves functionality with their respective flow.

4.3.1 Cal A Flow
4.3.2 Cal B Flow
4.3.3 Cal A/B IN Flow
4.3.4 Air Flow
4.3.5 Cal A/B/Air Flow
4.3.6 Cal C Flow

### 4.1 Reagent Priming

Go to **Menu** > **Diagnostics** > **Flow** > **Reagent Priming**.



**Diagnostics *Continued,***

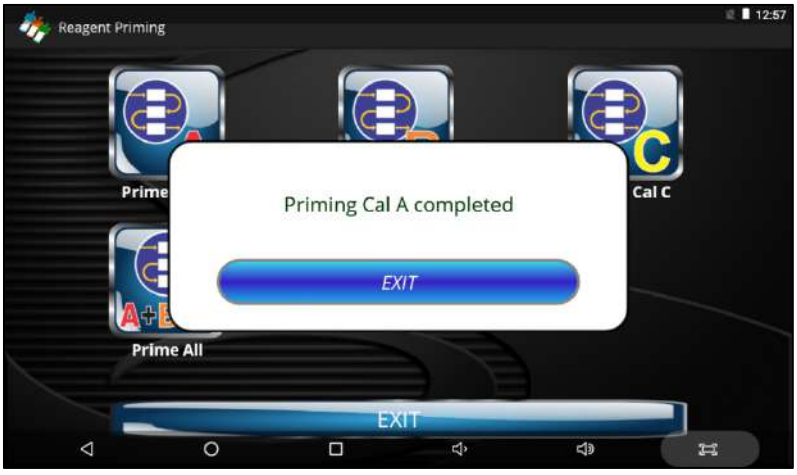
**Reagent Priming *Continued*,**

### 4.1.1 Prime Cal A

1. To perform Cal A priming sequence, press Prime Cal A.




2. After performing Cal A priming, priming gets completed. If instrument prompts the below screen or flags priming incomplete. Please check the tubing/reagent pack consumption for any errors.



**Diagnostics *Continued,***

**Reagent Priming *Continued*,**

### 4.1.2 Prime Cal B

1. To perform Cal B priming sequence, press Prime Cal B. After priming Cal B,instrument primes with Cal A for washing the electrodes.




2. After performing prime Cal B, priming gets completed. If Instrument prompts the below screen or flags priming incomplete. Please check the tubing/reagent pack consumption for any errors.



**Diagnostics *Continued,***

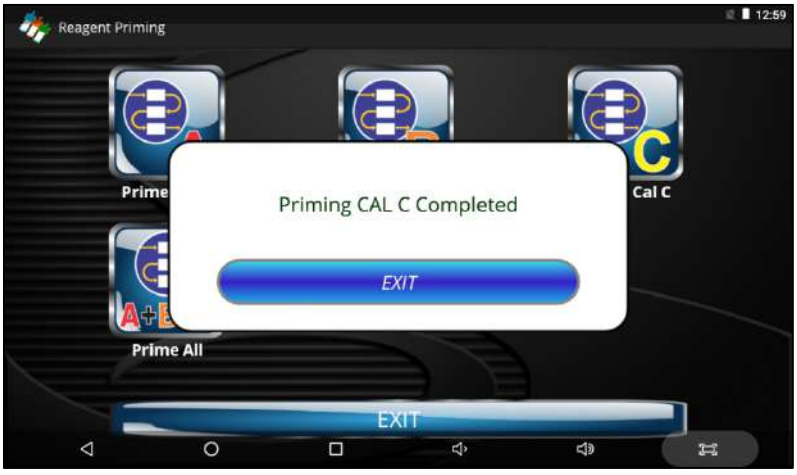
**Reagent Priming *Continued*,**

### 4.1.3 Prime Cal C

1. To perform Cal C priming sequence, press Prime Cal C.




2. After performing Cal C priming, priming gets completed. If instrument prompts the below screen or flags priming incomplete. please check the tubing/reagent pack consumption for any errors.



## Diagnostics *Continued,*

## Reagent Priming *Continued*,

### 4.1.4 Prime All

1. To perform Prime All sequence, press Prime All. After priming Cal B, instrument primes with Cal C and then Cal A .

**Priming Cal B**




**Priming Cal C**




**Priming Cal A**



**Diagnostics *Continued,***

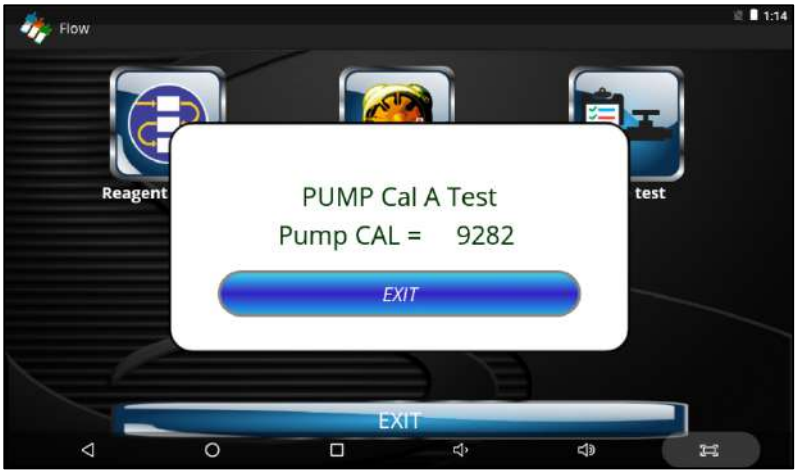
**Reagent Priming *Continued*,**

**4.1.4**
**Prime All**
***Continued,***




**4.2**
**Pump**
**Calibration**

1. Go to **Menu > Diagnostics > Flow > Pump Calibration.** Instrument performs pump calibration and displays its steps. Which defines the tubing durability.



**Diagnostics *Continued,***

## 4.3 Valve test

1. To perform valve test,

   Go to **Menu > Diagnostics > Flow > Valve test**




### 4.3.1 Cal A Flow

Allows user to test Cal A flow through valves as per its respective functionality. It activates Cal A flow and provides 300 seconds time duration to check its operation. When the time gets completed, it deactivates automatically or the user can off it by press the Stop button.



## Diagnostics *Continued,*

## Valve test, *Continues*

### 4.3.2 Cal B Flow

Allows user to test Cal B flow through valves as per its respective functionality. It activates Cal B flow and provides 300 seconds time duration to check its operation. When the time gets completed, it deactivates automatically or the user can off it by press the Stop button.




### 4.3.3 Cal C Flow

Allows user to test Cal C flow through valves as per its respective functionality. It activates Cal C flow and provides 300 seconds time duration to check its operation. When the time gets completed, it deactivates automatically or the user can off it by press the Stop button.



**Diagnostics *Continued,***

**Valve test, *Continues***

### 4.3.4 Cal A/B IN

Allows user to test Cal A/B IN flow through valves as per its respective functionality. It activates Cal A/B IN flow and provides 300 seconds time duration to check its operation. When the time gets completed, it deactivates automatically or the user can off it by press the Stop button.




### 4.3.5 Air Flow

Allows user to test Air flow through valves as per its respective functionality. It activates Air flow and provides 300 seconds time duration to check its operation. When the time gets completed, it deactivates automatically or the user can off it by press the Stop button.



**Diagnostics *Continued,***

**Valve test, *Continues***

**4.3.6 Cal A/B/Air Flow**

Allows user to test Cal A/B/Air flow through valves as per its respective functionality. It activates Cal A/B/Air flow and provides 300 seconds time duration to check its operation. When the time gets completed, it deactivates automatically or the user can off it by press the Stop button.



**Diagnostics *Continued,***

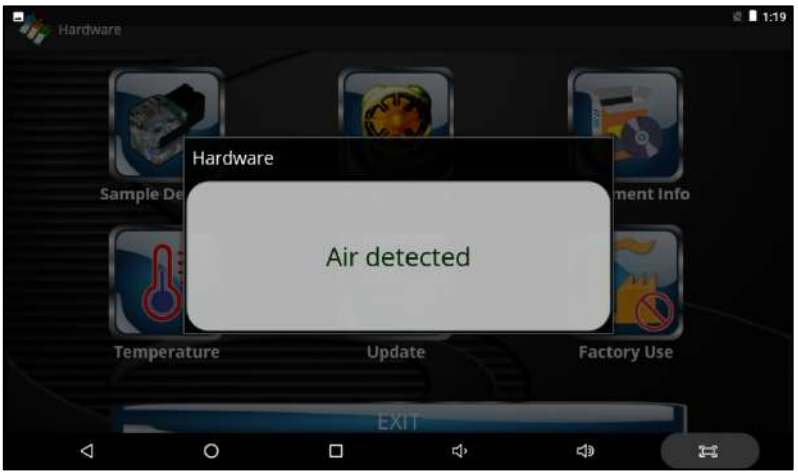
## 5. Hardware

It is used for checking the hardware functionality of modules used in the instrument.

5.1 Sample/bubble Detector
5.2 Pump Module
5.3 Instrument Info
5.4 Temperature
5.5 Update
5.6 Factory Use.

### 5.1 Sample/Bubble Detector

Allows user to test Sample/Bubble detector is working properly or not. If the sample detector looses its properties, user can check by this option. Initially, Instrument pump air, if it is detected. It pumps liquid, If it is detected., then sample/bubble detector is working fine. If either air/liquid is not detected, then instrument flags “Air not detected/liquid not detected” respectively. Need to replace it or troubleshoot it.






**Diagnostics *Continued,***

**5.2 Pump module**

Allows user to test pump motor functionality. Instrument rotates the pump with fast and slow sequences in order to identify if any error occurs during its operation.

Pump Speed : Fast sequence

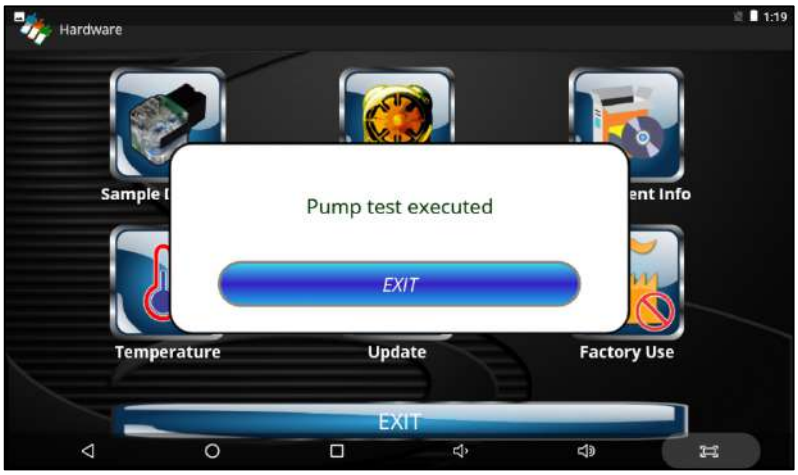



Pump Speed : Slow sequence




**5.2 Pump module**

If both speeds gets executed without any errors, then instrument prompts “pump test executed.”



**Diagnostics *Continued,***

**Pump module *Continued*,**

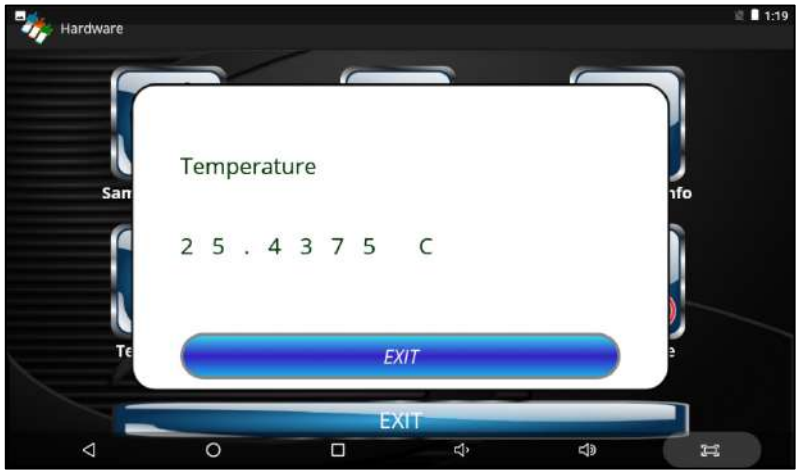
**5.3 Instrument Info**

Allows user to know the instrument serial number, software version and its development date.




**5.4 Temperature**

Allows user to know the working temperature of the instrument.




**5.5 Update**

Allows user to update to newer version. If there is any new version is updated in Sensa Core Domain. Then, it prompts to update to newer version.

**5.6 Factory Use**

Only for factory use.

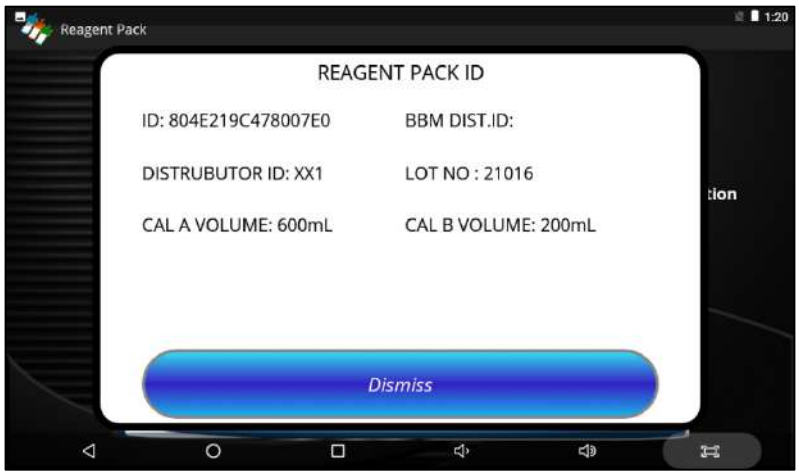
**Diagnostics *Continued,***

## 6. Reagent Pack

It shows the information of reagent pack like consumption, reagent pack lot no or pack volume & information about CAL A, CAL B and CAL C concentration.

### 6.1 Reagent Pack ID

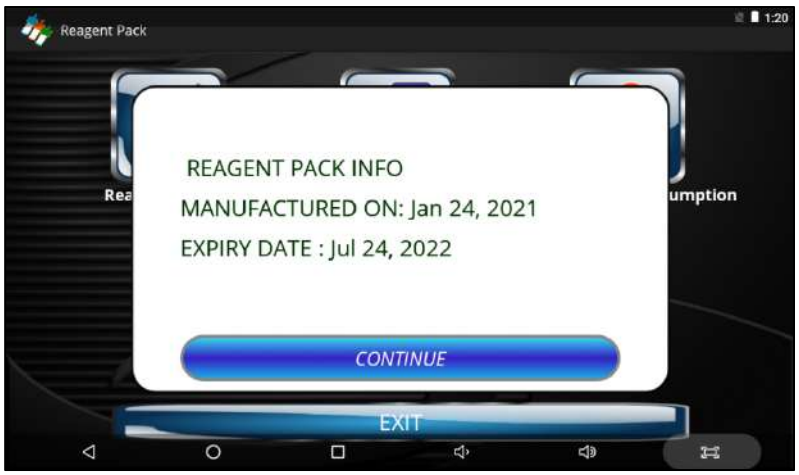
1. Go to **Menu > Diagnostics > Reagent Pack**
2. Select **Reagent Pack ID**, ***"Reagent pack ID Please wait"*** message will appear. Later the following screen will appear.




3. Press **Dismiss** to exit to the main screen.

### 6.2 Reagent Pack Info

1. Go to **Menu > Diagnostics > Reagent Pack**
2. Select **Reagent Pack Info**, ***"Reagent pack Info Please wait"*** message will appear. Later the following screen will appear.

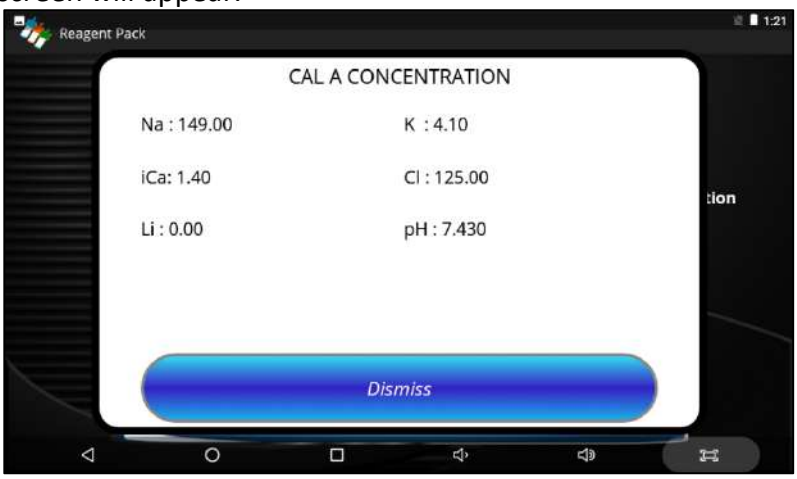


**Diagnostics *Continued,***

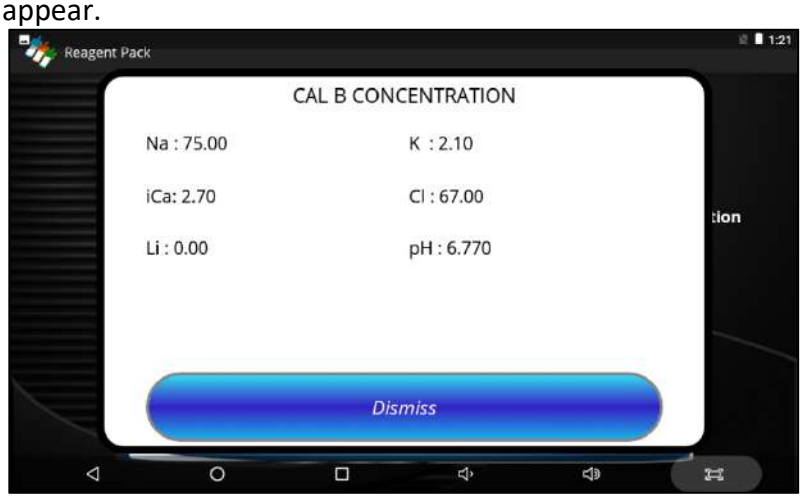
**Reagent Pack *Continued,***

**Reagent Pack Info *Continued,***

3. Press **continue** to know CAL A concentration. The following screen will appear.




4. Press **Dismiss** for Cal B concentration. The following screen will appear.




5. Press **Dismiss** for Cal C Concentration. The following screen will appear.

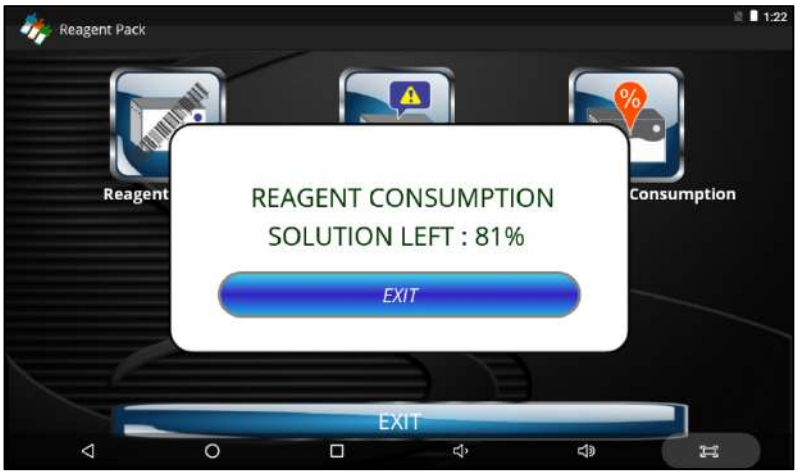


**Diagnostics *Continued,***

**Reagent Pack *Continued,***

**Reagent Pack Consumption**

1. Go to **Menu > Diagnostics > Reagent Pack**
2. Select **Reagent Pack Consumption**, ***"Reagent pack Consumption Please wait"*** message will appear. Later the following screen will appear.




3. Press **Exit**. The following screen will appear.




4. Press **Exit** to exit to the main screen.

## II. User Menu

**User Menu**

Go to **Menu > User Menu**

User Menu includes following options

**2.1 Normal Ranges**
- 2.1.1 Reset Ranges
- 2.1.2 User Settings
  - 2.1.2.1 Sample Ranges
  - 2.1.2.2 Urine Ranges
  - 2.1.2.3 QC1 Ranges
  - 2.1.2.4 QC2 Ranges
  - 2.1.2.5 QC3 Ranges

**2.2 Calibration Frequency**

**2.3 Sipping Frequency**

**2.4 Auto Standby**

**2.5 Correction Factor**

**2.6 Precision**



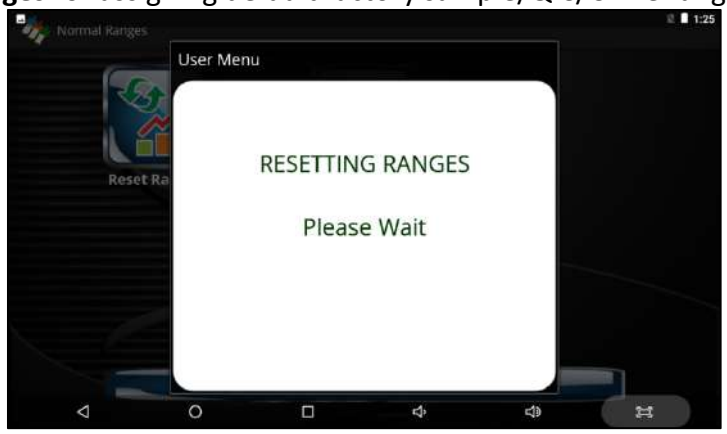
**User Menu *Continued,***

## 2.1 Normal Ranges

Go to **Menu > User Menu > Normal Ranges**
Used to set default or user defined sample/QC/Urine ranges.

### 2.1.1 Resetting ranges

1. Go to **Menu > User Menu > Normal Ranges > Resetting ranges** for assigning default factory sample/Q.C/Urine ranges.




2. Once the process is completed, instrument prompts Ranges Resetted.




### 2.1.2 User Settings

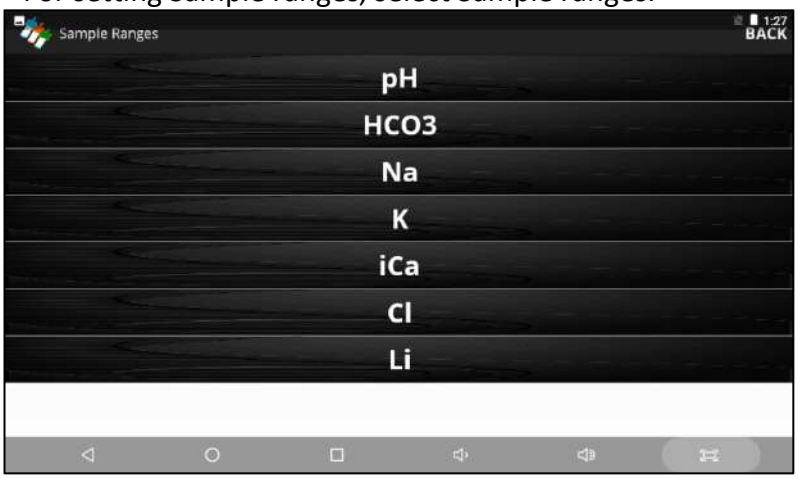
1. Go to **Menu**> **User Menu**> **Normal Ranges**> **User settings** For assigning user defined sample/Q.C/Urine ranges. Press User settings.



**User Setting *Continued,***

### 2.1.2 Sample Ranges

1. For setting Sample ranges, select Sample ranges.




2. Select respective parameter to enter its ranges. The following screen appears on the screen.




3. Press **Edit** to enter minimum and maximum limits of the sample range of selected parameter.



**User Setting *Continued,***

**2.1.2 User Settings *Continued,***

**Note:**

- ✓ Before saving the setup, the analyzer will validate the entered values.
- ✓ Patient Reference sample ranges should be established by individual institutions. The reference ranges in the table below are shown only as general guidelines.

| Parameter | Reference Ranges | Unit |
|---|---|---|
| pH | 7.33 -7.44 | N.A. |
| $pCO_2$ | 35.0 - 57.0 | mmHg |
| $Na^+$ | 135 -155 | mmol/L |
| $K^+$ | 3.5 - 5.5 | mmol/L |
| $iCa^{++}$ | 1.10 - 1.35 | mmol/L |
| | 4.4 - 5.4 | mg/dL |
| $Li^+$ | 0.30 - 1.50 | mmol/L |
| $Cl^-$ | 94 - 110 | mmol/L |
| $HCO_3^-$ | 22 - 28 | mmol/L |
| $TCO_2$ | 22.0 - 30.0 | mmol/L |
| BE | -3.0 - +3.0 | mmol/L |
| AGap_Na | 8.0 - 20.0 | mmol/L |
| AGap_K | 10.0 - 22.0 | mmol/L |

**User Setting *Continues,***

### 2.1.3 QC Ranges

1. Go to **Menu > User Menu > Normal Ranges > User Settings.**
2. Select **QC Ranges** (i.e., QC 1 Ranges, QC 2 Ranges, QC 3 Ranges). Parameter list appears on the screen.
3. Press on parameters and Dialogue box appears to edit the ranges.
4. Press edit to edit maximum and minimum values and Exit to main screen

### 2.1.4 Urine Ranges

1. Go to **Menu > User Menu > Normal Ranges > User settings**
2. Select **Urine Ranges,** Parameter list appears on the screen.
3. Press on parameter and Dialogue box appears to edit the ranges.
4. Press edit to edit maximum and minimum values and Exit to main screen

**Note:**

- ✓ Reference Urine ranges should be established by individual institutions. The reference ranges in the table below are shown only as general guidelines.

| Parameter | Unit | Reference Ranges |
|---|---|---|
| $Na^+$ | mmol/L | 40 - 220 |
| $K^+$ | mmol/L | 20 - 125 |
| $Cl^-$ | mmol/L | 100 - 250 |
| $iCa^{++}$ | mg/d | 100 - 300 |

**User Setting *Continues,***

**2.2 Calibration Frequency**

Go to **Menu**> **User Menu**> **Calibration Frequency**

Used to set auto calibration frequency to either 6 Hours or 12 Hours.




**2.3Sipping Frequency**

Go to **Menu**> **User Menu**> **Sipping Frequency**

Used to set Sipping frequency to either 1 Hour or 2 Hours.




**2.4 Auto Standby**

Go to **Menu**> **User Menu**> **Auto Standby**

Allows user to set auto-standby ON or OFF state. If auto standby is ON, instrument enters into standby mode after completing its auto calibration frequency. If it is in OFF state, then it won't enter into standby mode.

***Note :***

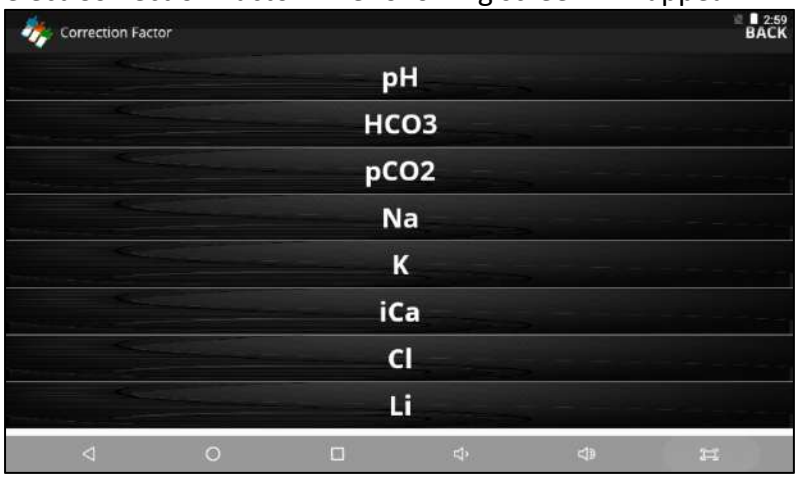
*In standby mode, instrument only performs sipping. It avoid auto calibration.*



**User Setting *Continues,***

**2.5 Correction Factor**

1. Go to **Menu > User menu**
2. Select Correction Factor. The following screen will appear.




3. Select desired parameter, the default values will appear.




4. Press set to Edit Slope/intercept values. The following screen appears.



**User Setting *Continues,***

**Correction Factor *Continued*,**

5. Press **SET** to enter the correction factor of the particular parameter.




6. Press **Save** to save the setup and go back to the main screen.

**Note:**

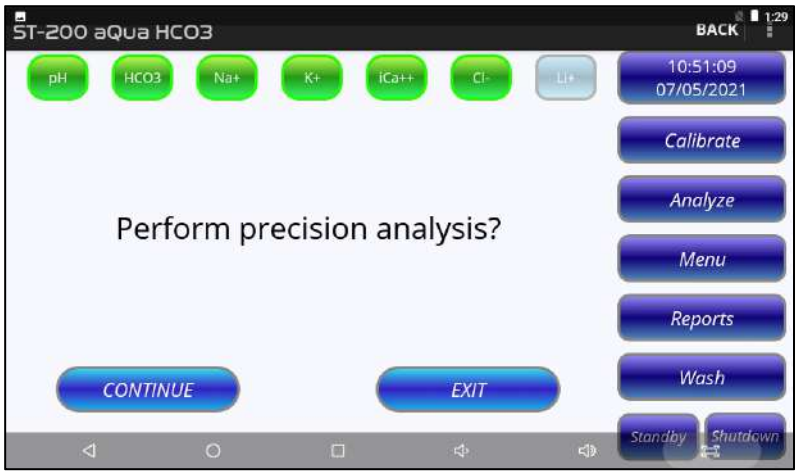
- ✓ Acceptable range for slope is from 0.5 to 1.5.
- ✓ Acceptable ranges for intercept are as following:

| Parameter | Intercept |
|---|---|
| $Na^+$ | -10.00 to +10.00 |
| $K^+$ | -0.50 to +0.50 |
| $iCa^{++}$ | -0.50 to +0.50 |
| $Cl^-$ | -10.00 to +10.00 |
| $Li^+$ | -0.50 to +0.50 |
| pH | -0.50 to +0.50 |
| $HCO_3^-$ | -10.00 to +10.00 |
| $pCO_2$ | -10.00 to +10.00 |

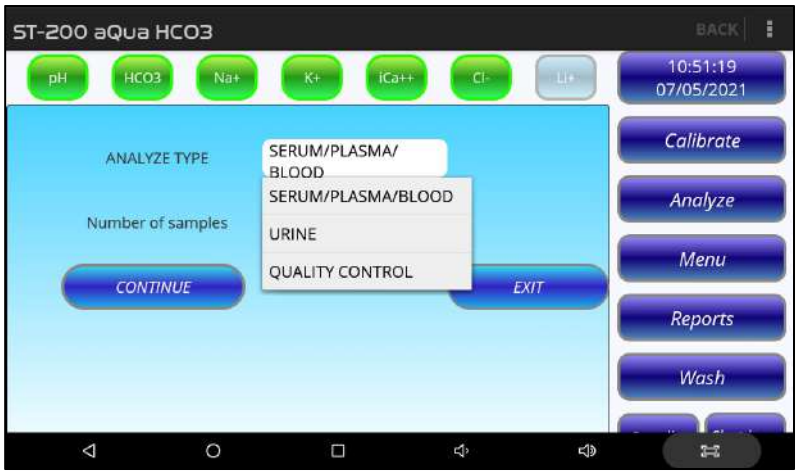
***User Setting Continued,***

## 2.6 Precision

1. Go to **Menu** on the main screen **> User Menu**
2. Select **Precision**. The following screen will appear.




3. Press **continue** to proceed. The following screen will appear.




4. Select the required criteria from the drop down list and press *Continue*. The following screen will appear according to the selected criteria.



**User Setting *Continued,***

**Precision *Continued,***

5. Press **continue**, the ***"Please Lift Sampler"*** message appears on the screen.
6. After lifting the sampler ***"Probe in Sample/QC/Urine"*** message appears on the screen.
7. Press **continue** to aspirate the sample ***"Please Wait"*** message appears on the screen.
8. After aspirating the sample ***"Please Remove Sample/QC/Urine & Close Sampler"*** message will appear.
9. After no. of samples, final precision result will be displayed on the screen.




***Note:***

- ✓ Analyze type for Precision results are as following:
  - Serum/Plasma/Blood
  - Urine
  - Quality Controls
- ✓ Select desired criteria in appropriate boxes.

## III. Setup Menu

### Setup Menu

1) Go to **Menu > Setup Menu**

**3.1 Analyze Type**

3.1.1 Serum/Plasma
3.1.2 Urine
3.1.3 Whole Blood
3.1.4 Quality Control
3.1.5 Micro Mode
3.1.6 Dilution

**3.2 Date and Time**

**3.3 Printer**

**3.4 Print-mV**

**3.5 Patient-ID**

**3.6 Print Parameter**



## Setup Menu *Continued,*

### 3.1 Analyze Type

Go to **Menu > Setup Menu>Analyze type**

Depending upon the samples to be aspired, user need to select respective analyze mode.




### 3.1.1 Serum/Plasma

Select Analyze type as Serum/Plasma for Serum/Plasma samples.




### 3.1.2 Urine

Select Analyze Type as Urine for Urine samples.



**Set up Menu *Continued,***

### 3.1.3 Whole Blood

Select Analyze Type as Whole Blood for Whole Blood samples.




### 3.1.4 QC

Select Analyze Type as Quality Control for QC samples.




### 3.1.5 Micro Mode

Select Analyze Type as Micro Mode for emergency results. The results will appear within 30 seconds.



**Setup Menu *Continued,***

### 3.1.6 Dilution Mode

Select Analyze Type as Dilution for Pediatric samples. Dilute 100 µL of Sample with 200 µL of CAL-A Solution (1:2 dilutions).



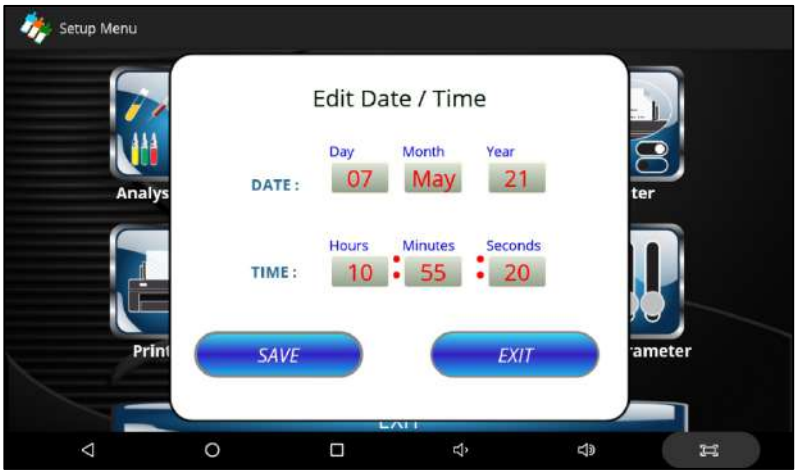
**Setup Menu *Continued,***

### 3.2 Date and Time

1. Go to **Menu > Setup Menu>Date and Time**
2. Select **Date and Time**. The following screen will appear.




3. Press **Set** button on the screen, following screen appears.




4. Press on Date format box and select the desired date, month and year format from the drop-down list.

5. Press on Time format box and select the desired time format (i.e., Hours, Minutes, Seconds) from drop down list.

6. Press **Save** to save the setup.

**Setup Menu *Continued,***

### 3.3 Printer

1. Go to **Menu > Setup Menu**
2. Select **Printer**. The following Screen will appear.




3. Select **ON** for printing the output data. Select **OFF** to not print the output results.

### 3.4 Print-mV

1. Go to **Menu > Setup Menu**
2. Press **Print-mV** the following screen appears.



**Setup Menu *Continued,***

## 3.5 Patient-ID

1. Go to **Menu > Setup Menu>Patient-ID**

Allows user to setup Patient ID ON/Manual or OFF. If it is in Manual, during every sample analysis instrument will prompt for patient ID, Name, sample No and operator ID as input. If it is in OFF state, it wont prompt any screen displaying patient details.

2. Select **Patient-ID.** The following screen will appear.




***Note:***

- ✓ If patient ID is in Manual mode, then database stores the data with respective patient fields. Else it stores no values in the patient details fields.

3. Press Exit to exit to the main screen.

**Setup Menu *Continued,***

## 3.6 Print Parameter

1. Go to **Menu > Setup Menu**
2. Select **Print Parameter.** The following screen will appear.




3. Select the parameter to print.




***Note:***

- ✓ If iCa parameter is selected, only iCa values are derived as output. No other parameters will be printed.
- ✓ If you want to print all parameters then select **ALL**.. All parameters will be printed.

4. Press **Exit** to exit to the main screen.

# 4. Sample Analysis

4.1 Patient Samples.....83

4.2 QC Samples..... 89

## 4.1 Patient Samples

### 4.1.1 Introduce Sample

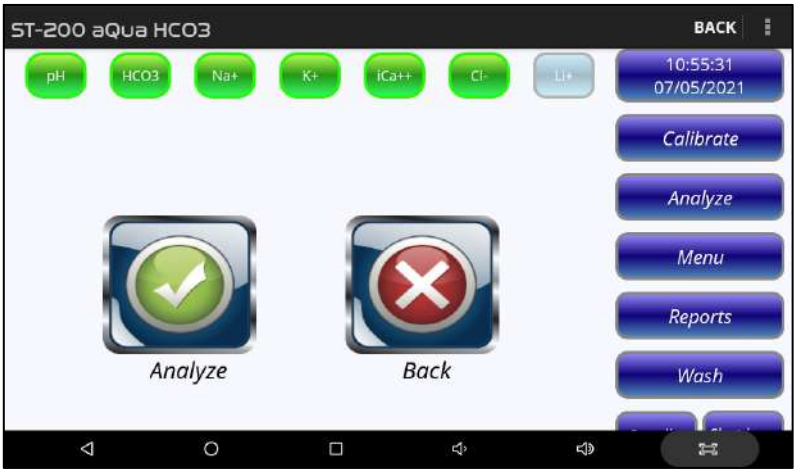
**Note:**

Sample Analysis is available only when the analyzer is in Ready State.

1. Check that the analyzer is in ready State. (**"Ready for Analysis").**
2. Now, analyze the sample by selecting the sample type.
3. Go to **Menu > Setup Menu > Analyze type**
4. Select the sample (i.e., Serum-Plasma, Urine, Whole blood, Quality Control, Micro mode and Dilution mode.




5. Press **Analyze** on the main menu screen to perform Analysis. The following screen appears.



**Patient Samples *Continued,***

**Introduce Sample *Continued,***

6. To continue press **Analyze**, the following screen appears.




7. Select the sample type [Serum/Plasma(Electrolytes) or Quality Control], the following screen appears on the screen.



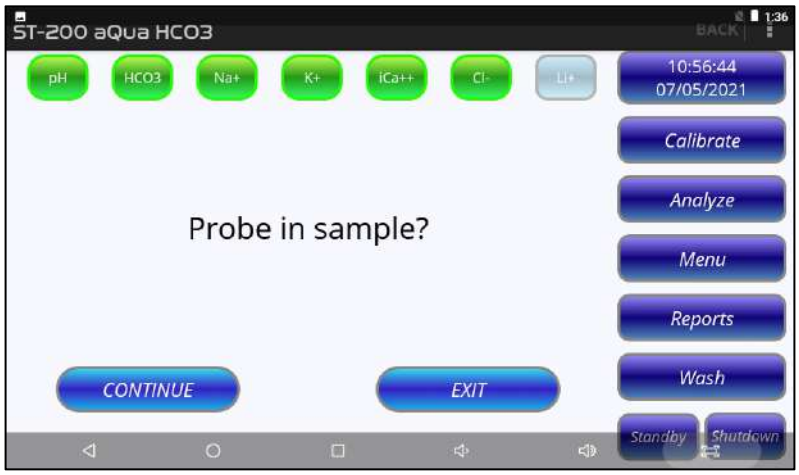

***Note:***

✓ Enter the details appeared on the screen using keypad.

**Patient Samples *Continued,***

**Introduce Sample *Continued,***

8. Lift up the sampler cover. The following message will appear.




*Press Continue to aspirate the sample*

***Note:***

- ✓ To cancel a sample, lower the sampler cover to its rest position and press exit.
- ✓ Once the continue option is pressed, the sample analysis process should not be disturbed. Wait until the screen display's result or asks required input's.

10. For the sample, immerse the end of the sampler probe into sample container.



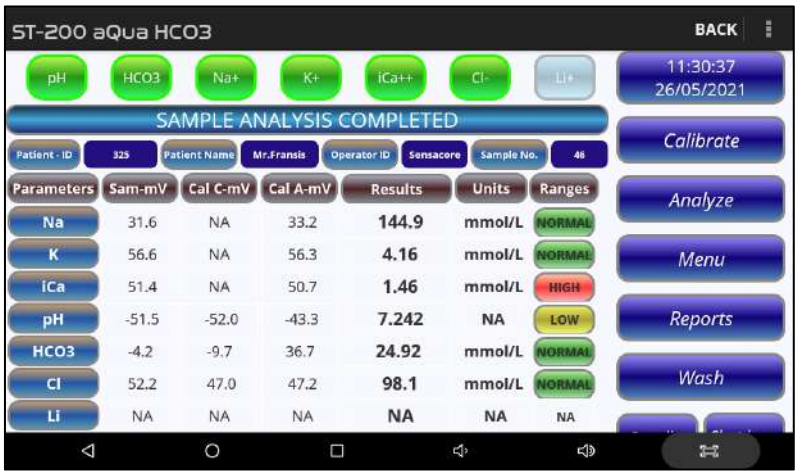
**Patient Samples *Continued,***

**Introduce Sample *Continued,***

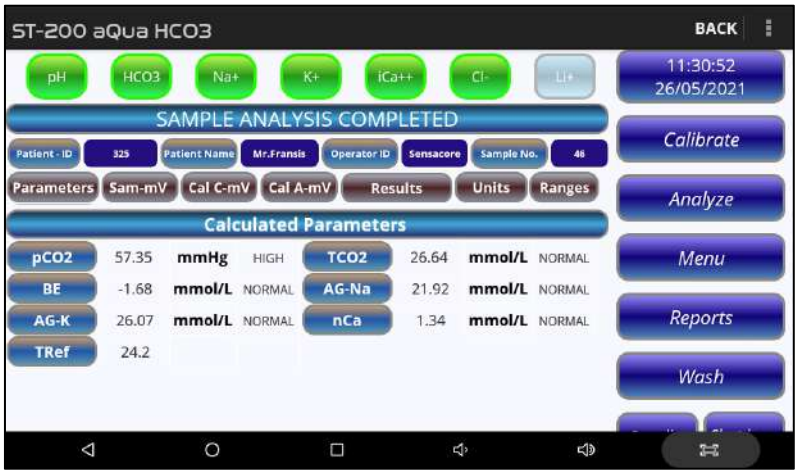
11. For a capillary sample, lift the sampler cover all the way up. Carefully insert the capillary into the probe.

**4.1.2 New Sample Results**

1. If the sampler probe is adequately immersed in the sample, press continue to begin aspiration.
2. Please wait message will appear on the screen while Aspirating.
3. If the aspiration is completed, ***"Please Remove Sample & Close Sampler"*** message appears.
4. Remove sample from sample probe and lower the sample cover to Close.
5. When the sample analysis is completed, the sample results will appear in the Results screen.




6. Scroll down for Metabolites and Calculated Parameter results.




7. The result values will appear in black color.

**Patient Samples *Continued,***

**New Sample Results *Continued,***

8. When the result is within the reference ranges, respective parameter ranges status will appear in **Green** color with **"Normal"** text.
9. When the result is above the reference range, respective parameter ranges status will appear in **Red** color with **"High"** text.
10. When the result is below the reference range, respective parameter ranges status will appear in **Yellow** color with **"Low"** text.
11. Error indicates the unknown value of the parameter.
12. If a sensor has failed, the corresponding result along with respective calculated parameter will not be reported.
13. The result of sample is printed based on the selected print status.

**Patient Samples *Continued,***
**New Sample**
**Results**
***Continued,***

```
ST-200AQUA
Patient ID : 325
Name       : Mr.Fransis
Operator ID: Sensacore
Sample No  : 46
- - - - - - - - - - - - - - -
---- SAMPLE mV ----
Na  =    31.6  mV
K   =    56.6  mV
iCa =    51.4  mV
pH  =   -51.5  mV
HCO3=    -4.2  mV
Cl  =    52.2  mV
-----CAL C mV-----
pH  =   -52.0  mV
HCO3=    -9.7  mV
Cl  =    47.0  mV
-----CAL A mV-----
Na  =    33.2  mV
K   =    56.3  mV
iCa =    50.7  mV
pH  =   -43.3  mV
HCO3=    36.7  mV
Cl  =    47.2  mV
TRef   = 24.2
---------------------
ANALYZE SAMPLE
---------------------
Na  =144.9 mmol/L
K   =4.16  mmol/L
iCa =1.46  mmol/L   HIGH
pH  =7.242 NA       LOW
HCO3=24.92 mmol/L
Cl  =98.1  mmol/L

----------------------
Normalized iCa at pH 7.4
----------------------
nCa =1.34 mmol/L
TCa =2.67 mmol/L
----------------------
CALCULATED PARAMETERS
----------------------
pCO2 = 57.35  mmHg    HIGH
TCO2 = 26.64  mmol/L
BE   = -1.68  mmol/L
Anion Gap
AG_Na= 21.92  mmol/L
AG_K = 26.07  mmol/L
----------------------
May-26-21   11:30:18
----------------------
Reagent:68% Samples:27
- - - - - - - - - - - - - -
```

Electrolytes

Calculated Parameters

## 4.2 QC Samples

### 4.2.1 Introduce QC Samples

1. Press **Menu > Setup Menu > Analyze Type** and select **Quality Control**.
2. If the quality control is selected for analysis, **"Analyze Type Quality Control"** message will appear.
3. Exit to the main screen and Press Analyze on main menu screen. The following screen will appear.




4. Press Analyze the following screen will appear on the screen




5. Enter the QC lot no., Sample No., and operator ID using Keypad. Select the QC Level from the drop down list and Press **Continue**.
6. When ***"Please Lift Sampler"*** displays, lift the sample cover. ***"Probe in QC?"*** message appears on the screen, insert QC sample in the probe and Press ***Continue.***
7. If the aspiration is completed, the ***"Please remove QC and close sampler"*** message will appear.
8. Remove QC sample from the sampler probe and push down the sampler cover to the original position.

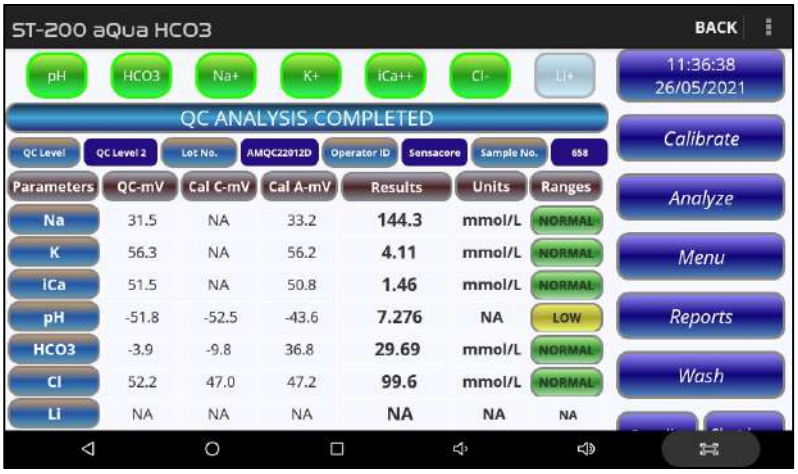
**QC Samples *Continued*, Introduce QC Samples *Continued*,**

***Note:***

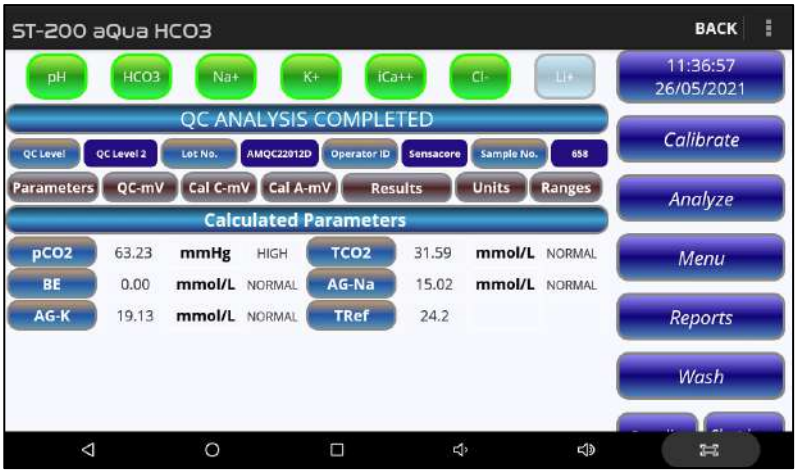
- ✓ To cancel QC sample, lower the sampler cover to its rest position before pressing Continue.
- ✓ Once Continue is selected, the sample analysis cannot be Canceled.

### 4.2.2 View QC Results

1. When the sample analysis is completed, the sample results will appear at the Results screen.



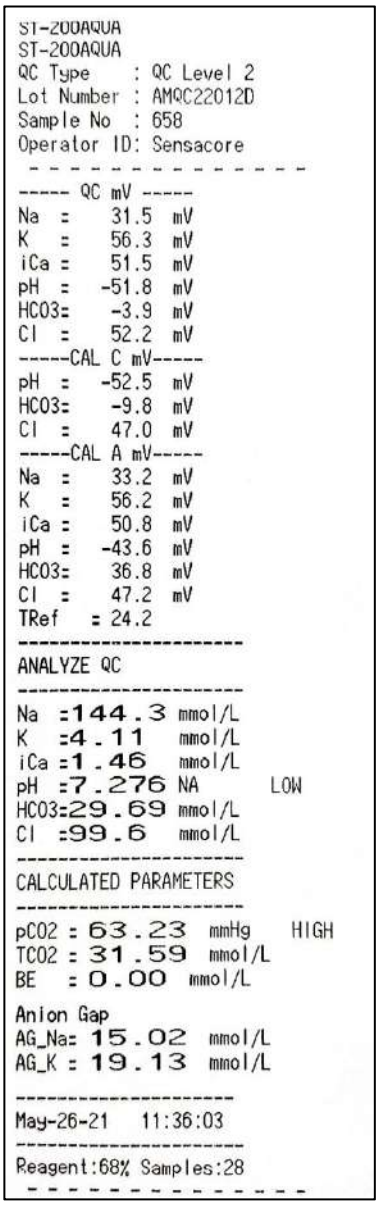

2. Scroll down for Calculated Parameter results.




3. The result values will appear in black color.
4. When the result is within the reference ranges, respective parameter ranges status will appear in **Green** color with **"Normal"** text.
5. When the result is above the reference range, respective parameter ranges status will appear in **Red** color with **"High"** text.
6. When the result is below the reference range, respective parameter ranges status will appear in **Yellow** color with **"Low"** text..

**QC Samples *Continued*, View QC Results *Continued*,**

7. If a sensor has failed the corresponding result will not be reported.
8. **"Error"** indicates the unknown value of the parameter or outside the measuring range.
9. The result of sample is printed based on the selected print status.



```
ST-200AQUA
ST-200AQUA
QC Type     : QC Level 2
Lot Number  : AMQC22012D
Sample No   : 658
Operator ID: Sensacore
 - - - - - - - - - - - - - - -
----- QC mV -----
Na  =    31.5  mV
K   =    56.3  mV
iCa =    51.5  mV
pH  =   -51.8  mV
HCO3=    -3.9  mV
Cl  =    52.2  mV
-----CAL C mV-----
pH  =   -52.5  mV
HCO3=    -9.8  mV
Cl  =    47.0  mV
-----CAL A mV-----
Na  =    33.2  mV
K   =    56.2  mV
iCa =    50.8  mV
pH  =   -43.6  mV
HCO3=    36.8  mV
Cl  =    47.2  mV
TRef   = 24.2
----------------------
ANALYZE QC
----------------------
Na  =144.3 mmol/L
K   =4.11  mmol/L
iCa =1.46  mmol/L
pH  =7.276 NA       LOW
HCO3=29.69 mmol/L
Cl  =99.6  mmol/L
----------------------
CALCULATED PARAMETERS
----------------------
pCO2 = 63.23  mmHg    HIGH
TCO2 = 31.59  mmol/L
BE   = 0.00   mmol/L

Anion Gap
AG_Na= 15.02  mmol/L
AG_K = 19.13  mmol/L

--------------------
May-26-21   11:36:03
----------------------
Reagent:68% Samples:28
 - - - - - - - - - - - - - - -
```

# 5. Data Base

5.1 Calibration Data..... 93
5.2 Patient (Sample/Urine) Data.....95
5.3 QC Data..... 98
5.4 Parameters.....100
5.5 L.J (Levey Jennings) Graph..... 101

## 5.1 Calibration Data

### 5.1.1 Calibration Results

1. Go to **Reports** on the Home screen.
2. Select **Calibration Results**, the search box appears on the screen and select the desired criteria from the drop down list.




3. If Date is selected from desired criteria the following screen will appear.



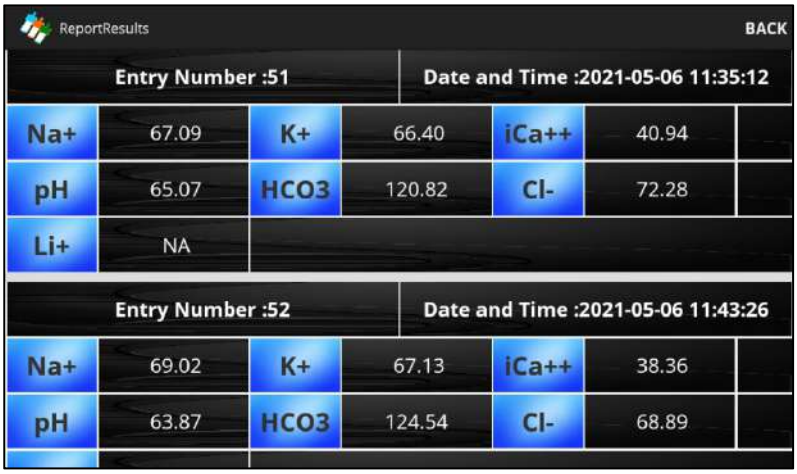

4. Enter starting and ending dates from the drop down list for the desired search period.



**Calibration Data *Continued,***

**Calibration Results *Continued,***

5. Press **Search**. The searched calibration results will appear as following.




***Note:***

- ✓ Search criteria for Calibration results are as following:
  - Row ID
  - Date
  - All Departments
- ✓ Enter desired search criteria in appropriate boxes.
- ✓ Press Search. The searched patient results will appear.
- ✓ Press back "⮌" button to exit the screen.

## 5.2 Patient (Sample/Urine) Data

Patient Data are stored on the analyzer and can be reviewed at any time. The following section will demonstrate how to find your data.

### 5.2.1 Patient Sample Results

1. Go to **Reports** on the Home screen. The following screen appears.




2. Select **Analyze Results**. Enter the desired search criteria in appropriate boxes.




3. Press **Search** for Patient results, the searched patient results will appear.



**Patient Data *Continued,***

**Patient Sample Results *Continued,***

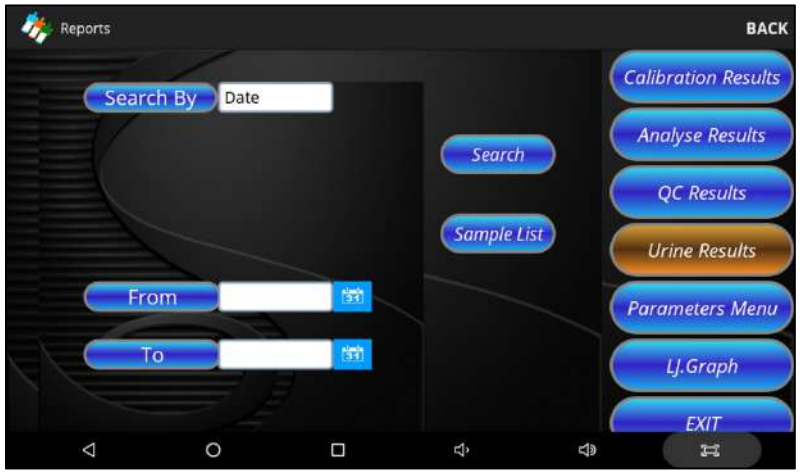
4. For printing the results, give a long press on the screen. A dialogue box appears on the screen to print. Press Print.
5. If there are more than one page of data, scroll the page with your finger to display all screens.

***Note:***

- ✓ Search criteria for patient results are as following:
  - Sample No.
  - Patient ID
  - Patient Name
  - Date
  - Operators ID
  - All Departments
- ✓ Enter desired search criteria in appropriate boxes.
- ✓ Press Search. The searched patient results will appear.
- ✓ Press back button to exit the screen.

**5.2.2 Patient Urine Sample Results**

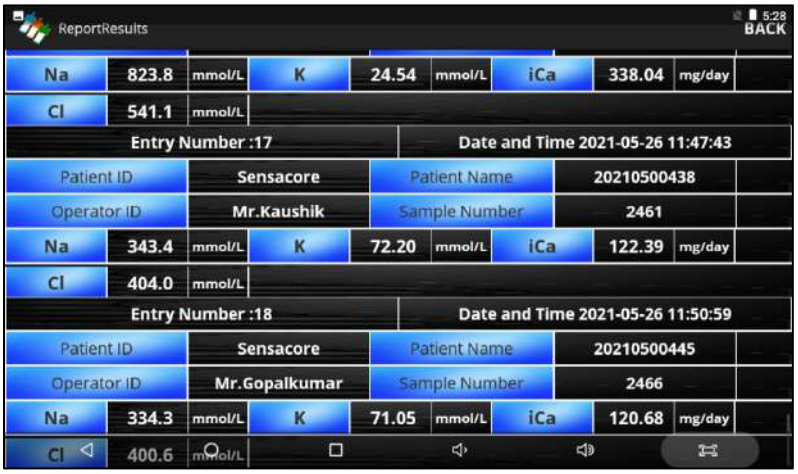
1. Go to **Reports** in the Main Menu screen and select **Urine Results**.
2. Enter the desired search criteria in appropriate boxes.



**Patient Data *Continued,***

**Patient Urine Sample Results *Continued,***

3. Press **Search** for Patient results, the searched patient results will appear.




4. For printing the results, give a long press on the screen. A dialogue box appears on the screen to print. Press Print.

***Note:***

- ✓ Search criteria for patient results are as following:
  - Sample No.
  - Patient ID
  - Patient Name
  - Date
  - Operators ID
  - All Departments
- ✓ Enter desired search criteria in appropriate boxes.
- ✓ Press Search. The searched patient results will appear.
- ✓ Press back "⮌" button to exit the screen.

## 5.3 QC Data

QC Data are stored on the analyzer and can be reviewed at any time. The following section will demonstrate how to find your data.

### 5.3.1 QC Results

1. Go to **Reports** in the main menu screen. Select **QC Results**, the search box appears on the screen.

2. Select the desired criteria from the drop down list and enter desired search criteria in appropriate boxes.




3. Press **search** for QC results. The following screen appears



**QC Data *Continued,***
**QC Results *Continued,***

4. For printing the results, give a long press on the screen. The dialogue box appears on the screen.




5. Select **Print** option to print the data.

***Note:***

- ✓ Search criteria for QC results are as following:
  - Lot No.
  - Operators ID
  - QC Level
  - Date
  - All Departments
- ✓ Enter desired search criteria in appropriate boxes.
- ✓ Press **Search**. The searched patient results will be displayed on the screen.
- ✓ Press back "⮌" button to exit the screen.

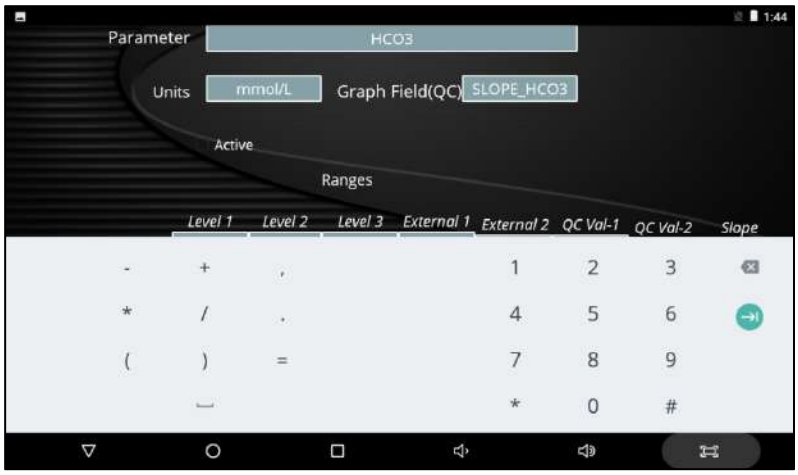
## 5.4 Parameters

### 5.4.1 Parameters Menu

1. Go to **Reports** and select **Parameters Menu**. The following screen will appear.




2. Select the desired parameter to change the slope. The following screen will appear.




3. Press **Edit icon** to edit the ranges.
4. After editing press **save icon** to save the changes.

***Note:***

- ✓ Level 1, Level 2, Level 3 are QC Levels
- ✓ External 1 and External 2 are Bio-rad Levels
- ✓ QC Val-1 and QC Val-2 are Cal A and Cal B

## 5.5 L.J (Levey Jennings) Graph

**LJ Graph**

1. To generate LJ graph on the analyzer proceed as follows, Go to **Reports** and select **LJ graph** to display the LJ graph Screen.

2. Select the desired parameter, QC levels and month from the drop down list for graph and press **Get Chart**. The following screen will appear.




3. **Exit** the screen by pressing back "⮌" button.
4. The graph consists of 3 sections

- The first **Yellow** section indicates the sample result is lower than the normal range - "Min Value".
- The middle **Green** section indicates the sample result is within the normal range - "Mean Value".
- The last **Red** section indicates the sample result is higher than the normal range - "Max Value".
- The **Blue** line represents the "Graph".

**LJ Graph *Continued,***

***Note:***

- ✓ The desired criteria in parameters are as follows
  - pH
  - $HCO^-_3$
  - $Na^+$
  - $K^+$
  - $iCa^{++}$
  - $Cl^-$
  - $Li^+$

- ✓ The desired criteria in QC Levels are as follows
  - QC Level 1
  - QC Level 2
  - QC Level 3
  - External Control 1
  - External Control 2
  - Self QC Level-1
  - Self QC Level-2

# 6. Standby & Shutdown

6.1 Standby Mode.....104
6.2 Analyzer Shutdown.....105

## 6.1 Standby Mode

**Standby Mode**

1. Go to **Standby** option on the Main menu.
2. ***"Entering the Standby Mode"*** message appears on the screen.
3. The following screen will appear when Standby mode is activated.




4. Press **Exit** button to exit stand by mode. The ***"Exiting Standby Mode"*** appears on the screen.

5. Later the following screen will appear.




6. Press **Exit** to exit to the main screen.

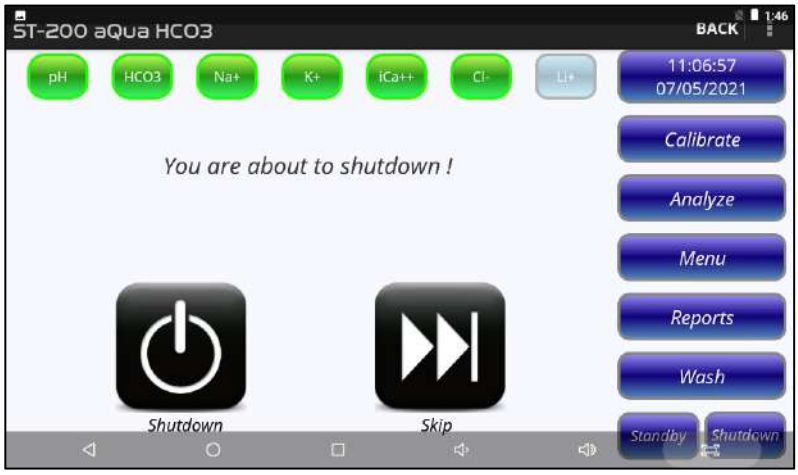
## 6.2 Analyzer Shutdown

**Caution**

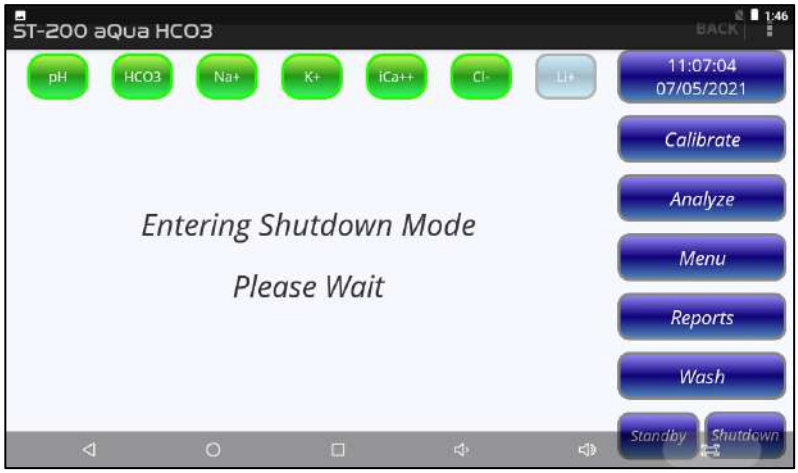
- ✓ The power of the analyzer should be turned off according to the shutdown procedure as described in this manual. Failure to follow the described shutdown procedure can cause damage to the data or the analyzer.
- ✓ To disconnect the power cables from the analyzer, first unplug the power cord from the outlet. Then, unplug the adapter from the analyzer.

### 6.2.1 Shutdown Analyzer

1. Go to **Shutdown** option on the main menu screen. The following screen will appear.




2. Press **shutdown icon.** The following message will appear.



**Analyzer Shut Down *Continued,***

**Shutdown Analyzer *Continued,***

3. Wait until the analyzer completes a power cycle and the following screen appears.




4. Facing the back of the analyzer, flip the red power switch on the lower left side, to the **"OFF"** position.

# 7. Maintenance

7.1 Maintenance Information.....108
7.2 Cleaning.....108

## 7.1 Maintenance Information

**Maintenance**

- All other maintenance is performed by replacing the components as described in the schedule below.
- When you wish to perform routine or any other maintenance function, go to the MENU option and Select “Maintenance”.
- The instrument removes all the liquid present in the tubing's by performing air priming.

## 7.2 Cleaning

**Caution**

- Wear appropriate personal protective clothing to prevent infection when cleaning.
- Do not spray cleaning solution directly onto the analyzer.
- Do not allow cleaning solution to enter the analyzer.
- Do not use force to wipe the screen.
- Dispose of all waste after cleaning in accordance with the
- laboratory’s established procedures for disposing of bio-hazardous materials.

**Cleaning Analyzer**

1. Use 0.5% hypochlorite cleaning solution.
2. Dampen a soft cloth with the cleaning solution.
3. Using a dampened soft cloth, wipe sampler cover, screen, and other contaminated areas on the analyzer.
4. Allow to air-dry for about 10 minutes.
5. Using a soft cloth dampened with water, wipe the analyzer.
6. Using a dry cloth, dry the surface of the analyzer.

**Cleaning the Touch Screen**

Immediately wipe any spilled liquid off of the touchscreen to prevent buildup of film on the screen. Clean the analyzer screen with a soft cloth dampened with isopropyl alcohol, a 10% bleach solution, or distilled water. Dry the screen with a soft cloth following cleaning.

***Caution:***

*Do not use strong detergents, concentrated bleach, or abrasive cleaning solutions that could scratch or damage the screen. Do not use excessive force.*

**Cleaning the Sample Probe**

Examine the probe outer and inner surface daily for dirt or contamination. Clean the probe surface using a swab moistened with isopropyl alcohol or a 10% bleach solution. The inner surface of the probe can be cleaned with spring needle if any blockage. Allow the probe to completely dry before testing.

**Cleaning the pinch valve flow path**

If any valve is blocked you can identify this by:

Go to Menu > Diagnostics > Flow > Valve Test

Flush the flow path with deionized water and re-calibrate the analyzer. Flushing the flow path removes the clots or other debris that may have been aspirated into the system.

# 8. Troubleshooting

8.1 Troubleshooting..... 111
8.2 Display Messages..... 120

## 8.1 Troubleshooting

This section describes the recommended troubleshooting procedures for use with ST-200 aQua Bicarbonate Analyzer. The procedures use the most logical and direct steps to resolve each problem and are written to minimize the replacement of any unnecessary parts.

**Guideline**

- If the problems described below are encountered during installation and/or operation of the analyzer, try the Suggested solutions as described in this manual.
- If the problem persists, please call a service engineer for further assistance.
- To enhance trouble-free operation, it is imperative to follow the maintenance schedule.When any error messages appear in the display, then refer to the list of "typical error messages" to locate the problem area and proceed to the appropriate section for troubleshooting instructions.
- *When performing troubleshooting procedures, put the analyzer in **MAINTAINANCE** mode.*

**A. Power**

If one of following situations applies*:*

- ✓ *The analyzer has turned off.*
- ✓ *The analyzer does not turn on.*

1. If not already done, turn off the power switch of the analyzer.
2. Check for loose power connections between the analyzer and the outlet. Tighten any loose power connections.
3. Turn on the power switch of the analyzer.
4. If the analyzer does not turn on, turn off the power switch of the analyzer.
5. Turn on the power switch of the analyzer again.

**B. USB Communication Error (Or) Tab Screen freeze**

Reopen the application and check it. If problem persists, restart the instrument or please call a service engineer for further assistance.

If one of following situations applies*:*

- *The screen does not respond.*
- *The screen is frozen.*
- *Abnormal screen appears.* Then,

1. Turn off the power switch of the analyzer.
2. Wait 10 seconds.
3. Turn on the power switch of the analyzer.

**Troubleshooting *Continued*,**

**C. Bar-code Scanner**

*When scanning bar-code, the scanner does not emit the red light:*

1. Remove the USB connector of the bar-code scanner from the analyzer.
2. Insert the USB connector into the same USB port of the analyzer. The bar-code scanner will make a beep sound and the indicator light on the scanner will turn on briefly.
3. Wait 5 seconds for the analyzer to recognize the bar-code scanner.
4. If the same problem occurs, repeat the procedure from step 1 with a different USB port of the analyzer.

*When scanning bar-code, the scanner emits red light, but does not take bar-code (no beep sound and no indicator light on):*

1. Position the bar-code scanner close to and in parallel with the Bar-code.
2. Check the bar-code for any damage. If damage is found, try again with another new Sample. Report the damaged Sample to a service engineer.

**D. Printer**

If one of following situations applies:

- ✓ *The printer does not print.*
- ✓ *The printer does not feed the paper.*

1. Open the printer cover.
2. Replace a roll of print paper if the paper has run out.

***Note:***

- ✓ The analyzer will resume printing in the printer queue if the printer has stopped.

**Troubleshooting *Continued*,**

**E. Flow Path System**

- All reagents and samples flow in one direction through the entire flow path system as shown. The flow path is a closed loop which ensures that the electrodes will always be kept wet, an important requirement of aQua $HCO_3$. All flow path connections are simple push-on, air tight connections, which requires no special tools.



## Troubleshooting *Continued*,

**Flow Path System *Continued,***

- If removal of ST-200 aQua $HCO_3$ Analyzer components are required during troubleshooting, the component removal order should be as follows:
  1) Electrode housing
  2) Sampler probe
  3) Valve module flow connections.

- For example, if any electrode requires replacement, it is necessary to go to MAINTAINANCE MENU for disconnecting the bubble detector tube from sampler and pump tube from the electrode housing. Then carefully remove the electrode housing.

- If the solutions valve requires cleaning, the operator must remove all of the items in the above list, before removing the solutions valve. To install components on the ST-200 aQua $HCO_3$, follow the installation order outlined under installation and replacement.

- Troubleshooting is covered in the order that problems are most likely to occur. Approach troubleshooting as a logical sequence of events. Isolate the problematic area to avoid unnecessary component replacement and/or excessive downtime.

**Troubleshooting *Continued,***

| **Flow Path System *Continued,*** | **PROBLEM AREA** | **TYPICAL ERROR MESSAGES** |
|---|---|---|
| | FLOW ERRORS | AIR IN CAL A,<br>AIR IN CAL B,<br>AIR IN CAL C,<br>AIR IN SAMPLE,<br>AIR IN CLEANER, & DRIFT. |
| | ELECTRODES | HIGH,<br>LOW,<br>DRIFT,<br>BUBBLE,<br>ERROR,<br>UNCALIBRATED. |
| | REAGENT PACKS | AIR IN CAL A,<br>AIR IN CAL B,<br>AIR IN CAL C,<br>INSUFFICIENT PACK<br>PLEASE REPLACE,<br>PACK EXPIRED,<br>INSERT PACK. |
| | SAMPLE/BUBBLE DETECTOR | AIR/LIQUID NOT-DETECTED,<br>AIR IN SAMPLE |
| | RFID | CONNECTION ERROR, INSERT PACK, INCOMPATIBLE PACK, PACK EXPIRED |

**Troubleshooting *Continued,***

**F. Flow Errors**

- Obstructions or air leaks in the flow path will result in error messages such as AIR IN CAL A, AIR IN CAL B, AIR IN CAL C, AIR IN SAMPLE & DRIFT. These messages indicate flow problems. A low or empty reagent pack may also cause AIR IN CAL A or AIR IN CAL B or AIR IN CAL C messages. If this problem exists, based on usage since the last reagent pack installation refer to reagent pack Consumption in Diagnostics Menu and troubleshooting before proceeding.

- If AIR IN CAL B, AIR IN CAL C or LIQUID NOT DETECTED! Message appears during a SOLUTION PRIMING or calibration, and reassembled, check to make sure that the electrodes are installed right side up and that all electrode connectors are properly installed. Performing this function twice may eliminate the error.

- FLOW ERROR (Valve fluidic error or Fluid path error): Flow error depends on following points.
  **1) Electrode housing**
  **2) Valves**
  **3) Reagent pack**

**Electrode Housing**

**Step 1:-** Remove sample detector tube from probe and attach it directly to pump tube which is connected at the bottom of reference housing. And check liquid flow as follows:

Press **Menu** > Press **Diagnostics** > Press **Flow** > Press **Reagent Priming** > Press **Prime Cal A**

Now, the pump rotates and liquid is aspirated through the tube. If **liquid is aspirated properly** follow step 2. If flow is not proper or no flow is observed. Kindly follow Step 3.

**Step 2:-** Then, there is no issue with valve system or flip seal or probe or tygon tube or pump tubing or reagent pack. Only module pending is reference housing electrode stack.

- Enter into maintenance mode and remove the electrode housing. Now, Clean deposited protein from electrodes with maintenance needle and reassembled electrode housing carefully.

- Reinsert the reference housing stag and check the flow. If problem gets resolved you can proceed with regular operation. Else please contact your concern service engineer.

**Troubleshooting *Continued,***

**Reagent Pack**

**Step 3:-**

The problem might be with reagent pack or valve system or probe seal or probe or tygon tube or pump tubing.

- **Reagent Pack :**

- Press Menu > Diagnostics > Reagent Pack > **Reagent Pack Consumption.**

- Check the reagent pack consumption. If it is above 20%, then remove the reagent pack and aspirate the liquid using syringe attached with tygon tube. If there is no flow observed, then replace the reagent pack and recheck the flow by repeating **Step 1**. Else if the flow is proper, then look into valve functionality and tubing's.

**Valve Test**

- **Valve System :**

- Press Menu> Diagnostics> Flow> **Valve test.** Check the valve system functionality individually as per its respective operation.
- **Valve 1(A/B):** It is used for aspiring calibrant solution from the reagent pack.

    1. While aspiring calibrant A valve 1(A/B) sets to A position and valve 2(Air) to OFF.
    2. While aspiring calibrant B valve 1(A/B) sets to B position and valve 2(Air) sets to OFF.

- **Valve 2(Air):** It is used for aspiring air then valve 2(Air) sets to ON.
- **Valve 3 (C):** It is used for aspirating Calibrant C Solution from the reagent pack.
    1. While aspirating Calibrant C valve 3 sets (A/B/Air) sets to OFF.
- If problem is observed in functionality, then check the connections on board. If problem still persists replace the pump motor driver card.
- Else if problem is not observed, check the valve tubing's if there is any damage or cut caused during valves operation. If any damage is found, replace it or contact your service engineer. If still the problem persists, check probe seal and tygon tubing.
- Check the probe seal and tygon tubing for any protein/salt blockage. If seal or tygon tube is damaged replace it. For further support contact your service engineer.

**Troubleshooting *Continued,***

**Electrode Errors**

- The electrode assembly should be replaced after 6 months of use. Sensa Core guarantees this item's performance for 1 month of use, but the frequency of replacement may depend on your sample volume or sample type (volume of lipemic high protein samples may necessitate more frequent changes of the electrode assembly). ST-200 aQua $HCO_3$ customers testing 100 or more patient samples per week should replace the electrode assembly after 6 months of use.

- The electrode assembly should be checked to prevent any leakage. At the same time, the electrode pins and pogo pins should be inspected to make sure they are dry and clean, including the reference electrode pin and Female connector, if low or erratic readings occur.

- Electrodes with low calibration values (below 35 for Na/K, below 30 for Li/pH/Cl/HCO3 & below 15 for iCa), Drift, or Noise messages may indicate electrode failure. Electrodes with HIGH calibration values messages most often indicate an electrode assembly/disposable reference electrode failure.

- Check the **Calibration Slope.** If they are within the following ranges.

  $Na^+$ : Should be in between 100 to 35
  $K^+$ : Should be in between 100 to 35
  $iCa^{++}$ : Should be in between 50 to 15
  $Cl^-$ : Should be in between 100 to 30
  $Li^+$ : Should be in between 100 to 30
  pH : Should be in between 100 to 30
  $HCO^-_3$ : Should be in between 130 to 30

  Then go for sample analysis or repeat calibration. After multiple calibration failure, replace the faulty electrode.

- For Na/K/iCa/Cl/Li/pH/$HCO_3$ analyzer, when the electrode housing is removed for electrode replacement, maintenance, troubleshooting, etc, wait for 10 minutes after re-installation to ensure proper electrode temperature.

  1. Calibration values (Slopes) of the electrodes should fall within a range mentioned above. If they fall outside of this range, the out-of-range values ******** for "Na", "pH", "K", "iCa", "Cl", "Li", "$HCO_3$" will be flagged. The analyzer should be recalibrated. At the end of the calibration cycle, the slope values will be displayed for each electrode. The ST-200 aQua $HCO_3$ will permit sample analysis only for the electrodes that are calibrated successfully.

**Troubleshooting *Continued,***

**Electrode Errors**

If Na or iCa or both Na & iCa not calibrated, Li won't analyzed, because Li Electrode depends on Na & iCa Electrodes.

2. A Na LOW, DRIFT message indicates that an additional Prime cycle may be necessary to condition the sodium electrode.

- If the calibration value remains low, replace the electrode.

3. If all electrodes have low or high calibration values, the problem is most likely with the electrode assembly or the reference electrode (on the Na/K/iCa/Cl/Li/pH/$HCO_3$ analyzer, the problem may be the disposable reference electrode). It is also possible that an incorrect reagent pack is being used, the ST-200 aQua $HCO_3$ will not calibrate successfully.

4. If drift messages appear for all electrodes, there is most likely a flow problem (refer to flow problems) or electrode assembly problem (refer to electrode assembly). Also, inspect the electrode pins, including the reference electrode pin, to ensure that they are dry and clean. After reassembly, the analyzer should be primed and calibrated.

- Do not perform the cleaning procedure described in paragraph **G** as a preventative measure. Do not clean the electrodes, as described, if their cal values are within the acceptable range. Do not clean the Potassium electrode with the sodium electrode brush tool under any circumstances.

5. In some cases, the chloride electrode calibration value may drop prematurely due to protein build-up on the inside surface of the chloride electrode. Attempt protein removal by pushing the brush tool (found in the troubleshooting Kit) through the chloride electrode. Gently pull the brush back out of the electrode. Reinstall the electrode, perform CALIBRATION. If the chloride electrode slope value remains low, install a new chloride electrode. Do not use the chloride brush to clean the sodium and potassium electrodes.

The bubble detector detects the difference between liquid and air, which is important for proper sample positioning in the ST-200 aQua $HCO_3$. To verify performance, Press> **Diagnostics** > **Hardware** > **Sample Detector**.

**G. Sample/ Bubble Detector**

In sample detector test, if liquid or air is not detected, clean Bubble Detector flow path with Sodium Hypo Chloride and DI water and perform sample detector test again. Then also if it is not detecting air or liquid replace the sample detector.
Finally install that bubble detector in instrument and perform sample detector test, it will work.

**Troubleshooting *Continued,***

## 8.2 Display Messages

| | |
|---|---|
| ANALYZE SAMPLE | : Perform Whole heparin blood, serum, CSF Sample analysis. |
| ANALYZE URINE | : Perform diluted urine Sample analysis. |
| ANALYZE QC | : Perform QC sample analysis. |
| AIR DETECTED | : Air detected in Sampler detector test. |
| AIR NOT DETECTED | : Air not detected in Sampler detector test. |
| AIR IN CLEANER | : Cleaning solution not detected correctly. |
| AIR IN SAMPLE | : Insufficient sample. |
| AIR IN CAL A | : Cal A not detected correctly. |
| AIR IN CAL B | : Cal B not detected correctly. |
| AIR IN CAL C | : Cal C is not detected correctly. |
| ASPIRATING | : Sample is being drawn into probe. |
| AUTO STANDBY | : Auto standby on/off |
| CALIBRATION FREQUENCY | : Set auto-calibration time. |
| SIPPING FREQUENCY | : Set auto-sipping time. |
| CALIBRATION SLOPE | : Display calibration values. |
| CALIBRATE | : Allows operator to calibrate analyzer. |
| CALIBRATING | : Analyzer is calibrating. |
| CAL A CONCENTRATION | : CAL A concentration values will be displayed. |
| CAL B CONCENTRATION | : CAL B concentration values will be displayed. |
| CAL C CONCENTRATION | : CAL C concentration values will be displayed. |
| SOLUTION LEFT | : Percentage reagent left in Reagent pack. |
| INCOMPATIBLE PACK | : Pack is not suitable for the instrument. |
| CLEANING… | : Analyzer is performing daily cleaning |
| CLEANING CYCLE PERFORMED … | : Appears when cleaning process is finished |
| SYSTEM UNCALIBRATED PLEASE CALIBRATE | : Instrument is not calibrated. Please Calibrate. |

## Troubleshooting *Continued,*

## Display Messages *Continued,*

| | |
|---|---|
| HIGH | : Result obtained are on the higher side. |
| LOW | : Result obtained are on the lower side. |
| DRIFT | : Result obtained are inappropriate due to mV variation as per standards. |
| ERROR | : Result obtained out of measuring ranges. |
| BUBBLE | : Result obtained is fault due to presence of air bubble in sample. Re-aspirate it. |
| DATE/TIME | : Allows operator to change date and time. |
| DAILY CLEANER | : Allows operator to perform daily cleaning. |
| DIAGNOSTICS | : Allows operator to check key analyzer functions. |
| DISTRIBUTOR ID | : Allows the operator to check distributors ID. |
| ENTERING STANDBY | : Analyzer will go into standby mode. Where only sipping is performed. |
| AIR IN SAMPLE | : Insufficient sample or Sample aspirated is not detected by sensor. |
| EXIT | : To clear the process or return to previous/idle screen. |
| PACK EXPIRED | : Check the expiry date of reagent pack and also check the date in instrument. |
| MANUFACTURERED ON | : Allows user to know the manufacturing date. |
| QC LEVEL 1<br>QC LEVEL 2<br>QC LEVEL 3<br>EXTERNAL QC 1<br>EXTERNAL QC 2<br>SELF QC LEVEL 1<br>SELF QC LEVEL 2 | : Quality control analyze mode.<br><br>**Note:**<br>Self QC levels performs analysis by aspirating Cal A and Cal B of reagent pack as sample. |
| LIQUID DETECTED | : Message will appear when the liquid is detected in sample detector test. |
| LIQUID NOT DETECTED | : Message will appear when the liquid is not detected in sample detector test. |
| LOT NO. | : Allows the operator to check lot number. |
| MENU | : Menu selection. |

**Troubleshooting *Continued,***

## Display Messages *Continued,*

| | |
|---|---|
| MAINTENANCE… | : Instrument is in maintenance mode. |
| POSITIONING SAMPLE | : Message will appear during sample positioning. |
| PROBE IN SAMPLE? | : Request operator to confirm whether probe is in sample. |
| PLEASE INSERT PACK | : Message will appear when the system has not detected reagent pack. |
| PRIMING CAL A | : Cal A fluid is being primed. |
| PRIMING CAL B | : Cal B fluid is being primed. |
| PRIMING CAL C | : Cal C fluid is being primed. |
| PRIME ALL | : Both Cal B, Cal C and Cal A fluids/calibrant's are primed. |
| PRINTER | : Allows user to on the printer or of the printer. |
| PRINT-mV | : Allows user to print the results with mV. |
| PRINT REFERENCE RANGES | : Allows user to print the reference ranges. |
| PROBE IN CLEANER | : Requests operator to confirm that probe is in Daily Cleaning Solution. |
| PUMPING AIR | : Message will appear when air pumps |
| PUMPING CAL A | : Message will appear when CAL A reagent get pumped. |
| PUMPING CAL B | : Message will appear when CAL B reagent get pumped. |
| PUMPING CAL C | : Message will appear when CAL C reagent get pumped. |
| PUMP CAL ERROR | : Failure of Pump cal Test. |
| PUMPING LIQUID | : Message will appear when liquid is pumped. |
| PUMP SPEED : FAST | : Allows the operator to set the pumping speed fast. |
| PUMP SPEED : SLOW | : Allows the operator to set the pumping speed slow. |
| REAGENT CONSUMPTION | : Allows the operator to check the quantity of the reagent solution consumed. |
| REAGENT PACK | : Allows operator to check the reagent pack details. |

**Troubleshooting *Continued,***

## Display Messages *Continued,*

| | |
|---|---|
| REAGENT PACK CONSUMPTION | : Allows operator to display % pack expended. |
| REGEANT PACK ID | : Allows the operator to check the pack ID. |
| REAGENT PACK INFO | : Allows the operator to check the pack information. |
| REMOVE SAMPLE FROM PROBE | : Requests operator to remove sample from probe. |
| SAMPLE DETECTOR TEST | : Message will appear during sample detect test |
| NORMAL RANGES | : Used to set the Na/K/iCa/Cl/Li/pH/$HCO_3$ sample limits. |
| ANALYZE TYPE | : Allows operator to select the sample type. Eg: Serum-Plasma, Urine, Whole Blood, QC |
| ST-200AQUA | : Instrument model name. |
| SENSOR CAL A - mV | : Allows the operator to display the mV readings of CAL A reagent selected. |
| SENSOR CAL B - mV | : Allows the operator to display the mV readings of CAL B reagent selected. |
| SENSOR CAL C - mV | : Allows the operator to display the mV readings of CAL C reagent selected. |
| STAND BY | : Appears when Analyzer is in Standby mode. |
| PACK SAMPLE COUNTER | : Allows the operator to check the total samples performed with the pack. |
| DAILY CLEANER COUNTER | : Allows the operator to check the total Daily Cleaners performed with the pack. |
| CUMULATIVE SAMPLE COUNTER | : Allows the operator to check the total samples performed in the instrument. |

# 9. Specifications

9.1 Operating Specifications..... 125
9.2 Instrument Specifications..... 126
9.3 Reagent pack Specifications.....127
9.4 Performance Specifications..... 128
9.5 Principles of Operation..... 129
9.6 Calculated Parameters.....130

## 9.1 Operating Specifications

**Measured Parameters**

- Measured Parameters Ranges Resolution

**Serum/Whole Blood:**

| Parameter | Measurement Range | Reproducibility (CV%) |
|---|---|---|
| $Na^+$ | 20.0 - 250.0 mmol/L | < 1.0 % |
| $K^+$ | 0.20 - 40.0 mmol/ L | <2.0% |
| $iCa^{++}$ | 0 - 10.0 mmol/L | <2.0% |
| pH | 5 - 9 | ± 0.03 |
| $HCO^-_3$ | 5.0 - 50.0 mmol/L | <3.0% |
| $Cl^-$ | 20.0 - 200.0 mmol/L | <1.0% |
| $Li^+$ | 0 - 10.0 mmol/L | <2.0% |

**Urine :**

| Parameter | Measurement Range | Reproducibility (CV%) |
|---|---|---|
| $Na^+$ | 25.0 - 1000.0 mmol/L | <5.0% |
| $K^+$ | 1.0 - 500.0 mmol/L | <5.0% |
| $Cl^-$ | 25.0 - 500.0 mmol/L | <5.0% |
| $iCa^{++}$ | 50.0 - 1000.0 mg/d | <5.0% |

- Sample types: Whole Blood, Serum, Plasma, CSF and Diluted Urine
- Sample volume: 120 µL
- Sample introduction method: Aspiration
- Reading time: 90 seconds

**Operating Specifications *Continued,***

**Environmental Conditions**

- Measuring principle: Direct measurement with ion-selective electrode (ISE).
- Calibration : Automatic or Manual
- Operating location: Indoors on a flat surface
- Operating temperature: 10 °C - 35 °C, <85% moisture
- Power supply: 100 - 240 VAC, 50/60 Hz (± 10%)

## 9.2 Instrument Specifications

**Instrument Specifications**

- Operating System: Android Nougat OS
- Display: 7 inch high definition LCD with capacitive touch
- Printer: 2" thermal printer (internal printer)
- Printer paper: Thermal (2" inch 24 column)
- Port: USB (2 ports), Serial (RS-232)
- Power adapter:

  - AC-DC power supplies

  - Input: 100/115-V AC, 50 - 60Hz or 220 VAC, 50/60 Hz, 0.75Amp

  - Output: 12 V, 5.0A, 60 W max
- Storage temperature: 10 - 35 °C
- Size (HxWxD): 15.2" W x 13.2" H x 8.7" D
- Weight: 4.3 kg

## 9.3 Reagent pack Specifications

**Reagent Pack**

- Unit: 1 Reagent Pack
- Shelf life: 18 months from manufactured date
- Storage temperature: 18 °C - 25 °C
- Components:
  - Waste bag (Biohazard after use)
  - Cal A solution, approx. 600 mL
  - Cal B solution, approx. 200 mL
  - Cal C Solution, approx. 350 mL

- Composition of calibration solutions

CAL A : Electrolytes, Buffers and Preservatives

CAL B : Electrolytes, Buffers and Preservatives

CAL C : Electrolytes, Buffers and Preservatives

## 9.4 Performance Specifications

### 9.4.1 Introduction

The results obtained during evaluation of ST-200 aQua $HCO_3$ Analyzers at Sensa Core are shown below to demonstrate the typical performance characteristics of the analyzer.

### 9.4.2 QC Precision

Precision of an ST-200 aQua $HCO_3$ Analyzer were obtained using Sensa Core QC levels .

| Sensa Core QC Level 1 | | | | | | |
|---|---|---|---|---|---|---|
| **Parameters** | **$Na^+$** | **$K^+$** | **$iCa^{++}$** | **pH** | **$HCO_3^-$** | **$Cl^-$** |
| 1 | 126.4 | 2.1 | 2.18 | 7.107 | 27.45 | 77.1 |
| 2 | 125.4 | 2.06 | 2.16 | 7.112 | 27.69 | 76.6 |
| 3 | 125.3 | 2.07 | 2.17 | 7.12 | 28.55 | 77.4 |
| **Mean** | **125.7** | **2.077** | **2.17** | **7.11** | **27.90** | **77.03** |
| **STD** | **0.6** | **0.021** | **0.01** | **0.01** | **0.58** | **0.40** |
| **CV** | **0.50** | **1.00** | **0.46** | **0.09** | **2.07** | **0.52** |

| Sensa Core QC Level 2 | | | | | | |
|---|---|---|---|---|---|---|
| **Parameters** | **$Na^+$** | **$K^+$** | **$iCa^{++}$** | **pH** | **$HCO_3^-$** | **$Cl^-$** |
| 1 | 144.0 | 4.05 | 1.44 | 7.291 | 30.1 | 99.4 |
| 2 | 144.2 | 4.14 | 1.45 | 7.296 | 30.28 | 99.5 |
| 3 | 144.1 | 4.17 | 1.44 | 7.298 | 30.02 | 100.1 |
| **Mean** | **144.10** | **4.12** | **1.44** | **7.295** | **30.13** | **99.67** |
| **STD** | **0.10** | **0.06** | **0.01** | **0.004** | **0.13** | **0.38** |
| **CV** | **0.07** | **1.52** | **0.40** | **0.04** | **0.44** | **0.38** |

| Sensa Core QC Level 3 | | | | | | |
|---|---|---|---|---|---|---|
| **Parameters** | **$Na^+$** | **$K^+$** | **$iCa^{++}$** | **pH** | **$HCO_3^-$** | **$Cl^-$** |
| 1 | 166.0 | 6.04 | 0.88 | 7.498 | 33.26 | 114.3 |
| 2 | 166.2 | 6.01 | 0.91 | 7.5 | 31.93 | 114.1 |
| 3 | 166.4 | 6.08 | 0.87 | 7.508 | 31.54 | 113.9 |
| **Mean** | **166.20** | **6.04** | **0.89** | **7.50** | **32.24** | **114.10** |
| **STD** | **0.20** | **0.04** | **0.02** | **0.01** | **0.90** | **0.20** |
| **CV** | **0.12** | **0.58** | **2.35** | **0.07** | **2.80** | **0.18** |

## 9.5. Principles of Operation

### 9.5.1. ISE Theory

Electrolyte measurements in blood products were traditionally performed using flame photometry, in which a sample, diluted with a known concentration of a reference ion (usually lithium or cesium), is aerosolized and passed through a flame that excites the cation. They re-emit the energy as light of different frequencies; the amplitude of this emission is proportional to the ion concentration in the sample. The development of sodium and pH-selective glass, and selective organic compounds for potassium and chloride, has permitted the development of sensors capable of measuring biological fluids directly, throughout the physiological range. These sensors are known as Ion-selective electrodes.

The ST-200aQua HCO3 electrolyte analyzer measures sodium, potassium, ionized calcium, lithium and chloride in biological fluids, using ion selective electrode technology. The flow-through sodium electrodes contain plasma tube, specially formulated to be sensitive to sodium ions. The flow-through potassium electrodes employ a plastic tube, incorporating neutral carrier ionophores. The flow-through Ionized Calcium electrodes includes a plastic tube, specially formulated to be selective to Calcium ions. The flow-through Lithium electrodes includes a plastic tube, specially formulated to be selective to Lithium ions. The flow-through Chloride electrodes includes a plastic tube, specially formulated to be selective to Chloride ions. The flow-through pH electrode includes a plastic tube, specially formulated to be selective to Hydrogen ions. The potential of each electrode is measured relative to a fixed, stable voltage established by the silver/silver chloride reference electrode. An ion selective electrode develops a voltage that varies with the concentration of the ion to which it responds. The relationship between the voltage developed and the concentration of the sensed ion is logarithmic, as expressed by the Nernst equation:

$$E = E^{\circ} + \frac{RT}{nF} \text{Log} (a_C)$$

Where:

| | | |
|---|---|---|
| E | = | the potential of the electrode in sample solution |
| Eº | = | the potential developed under standard conditions |
| RT/nF | = | A temperature dependent “constant” termed the slope (s) |
| n | = | 1 for sodium, potassium, chloride, lithium and pH |
| n | = | 2 for calcium |
| Log | = | Base ten logarithm function |
| a | = | Activity coefficient of the measured ion in the solution |
| c | = | Concentration of the measured ion in the solution |

A comparative method of measurement is utilized. First, the analyzer measures the potential developed when the sample is aspirated through the electrodes. Next, cal A is aspirated through the electrodes.

The difference in the two potentials is related logarithmically to the concentration of ions in the sample divided by their respective concentrations in the calibrant solution. Since the difference in potentials and the concentration of the ions in the calibrant solution are known, the analyzer can calculate the concentration of the ions in the sample solution, in accordance with the Nernst equation, rewritten as:

$$E\text{-}E^{o} = S \log (Ci(x) / Ci(s)) \text{ or } Ci(x) = Ci(s) \times 10^{\wedge} (E\text{-}E^{o})/ S$$

Where: E = ISE potential developed in sample solution
Eº = ISE potential developed in the calibrant solution
S = Electrode Slope calculated during calibration
Ci(x) = Concentration of ion "I" in the sample
Ci(s) = Concentration of ion "I" in the calibrant solution

"S", the slope, is determined during calibration using Cal A and B, which are known levels of ions.

The ST-200aQua HCO3 electrolyte analyzer aspirates and measures Cal A, followed by Cal B. The ST-200aQua HCO3 electrolyte analyzer finally aspirates an additional Cal "A" solution and measures it. The two Cal "A" readings are compared for drift and each measurement is checked for instability (noise). The slope is calculated from the difference between the second Cal A reading and the Cal B reading. A slope outside the built-in software limits will be flagged and displayed as low or high. The ST-200aQua HCO3 electrolyte analyzer will detect excessive drift or noise and display the appropriate error message.

## 9.6 Calculated Parameters

**AG_Na = Anion Gap**
Formulae used to calculate AG is given below
AG (Na+) = (Na result) – (Cl Result) – ($HCO_3$ Result)

**AG _K = Anion Gap for K**
Formulae used to calculate AG (K+) is given below
AG (K+) = (Na result) + (K result) – ($HCO_3$ Result) – (Cl Result)

**BE = Base Excess**
Formulae used to calculate BE is given below
BE = (0.93* $HCO_3$ Result) + (13.77*pH Result)-124.58

**pCO2 = Partial Pressure of Carbon dioxide**
Formulae used to calculate pCO2 is given below
pCO2 = (HCO3 result) * {exp [(7.604 - pH result)] * 2.303}

**TCO2 = Total $C0_2$**
Formulae used to calculate TCO2 is given below
TCO2 = ($HCO_3$ Result) + (0.03*pCO2)

**nCa = Normalized Calcium**
Formulae used to calculate nCa is given below
nCa = {iCa result * [(-0.24) * (7.4 - pH result)] * 2.303} + (iCa result)

**TCa = Total Calcium**
Formulae used to calculate TCa is given below
TCa = (nCa result) * 2

# Appendices

## Appendix A : Sample Collection and Handling

**Caution**

- Make sure there are no small air bubbles trapped in the collected sample. Small bubbles in the sample can affect the results.
- Do not use liquid anticoagulants as they can dilute the sample and affect the results.
- Hemolysis at any stage of sample preparation may cause erroneously high potassium results.
- If blood is not mixed completely with anticoagulant, blood clots can be partially formed in the sample. Samples containing clots must not be used as clots will cause sampling failure in the cartridge.
- Samples in which hemolysis is present or suspected and samples collected more than one hour before analysis must not be used.
- Use heparinized vacuum blood collection tubes for whole blood.
- Collect blood up to the fill indicator on the tube label.

**Whole Blood**

- Immediately after collecting blood, roll the tube between palms or gently shake it slowly up and down 8 to 10 times to thoroughly mix he blood and anticoagulant.

**Serum**

- Use non-heparinized, plain vacuum blood collection tubes for serum separation.
- Collect blood up to the fill indicator on the tube label.
- After collection, gently shake the tube slowly up and down five times.
- Allow the blood collection tube to stand vertically for about 30 minutes to fully clot before centrifugation.
- Centrifuge the blood sample, then collect and transfer the serum (liquid component separated from solid components of blood) to another clean container.

## Appendix A : Sample Collection and Handling, *continued*

### Plasma

- Use heparinized vacuum blood collection
- n tubes for plasma separation.
- Collect blood up to the fill indicator on the label.
- Roll the tube between palms or gently shake the tube slowly up and down 8 to 10 times.
- Centrifuge the tube immediately after collection.
- Collect and transfer the plasma (liquid component separated from solid components of blood) to another clean container.

## Appendix B :

## Warranty

The Product warranty for the analyzer will be free of defects in material and workmanship for a period of twelve months from the date of purchase, except for those parts or materials which are consumed or expended in normal use.

In the case of consumables or expendables, the warranty time period shall be used within the time prescribed providing such consumable or expendable is used prior to any “install by” or “expiration date”.

The warranty is only applicable for ST-200 aQua $HCO_3$ Analyzer under normal operating conditions and maintained in accordance with the instructions in the operator’s manual and when not having been subjected to accident, alteration, or misuse.

In the event of failure within the warranty period, Sensa Core will repair or replace any Analyzer returned. Sensa Core reserves the right to ask for proof of date of purchase, such as original invoice and / or maintenance records.

Representations and warranties made by any person including dealers, representatives and employees, which are inconsistent or in conflict with, or in addition to, the terms of this warranty shall not be binding upon Sensa Core unless in writing and signed by one of its officers.

* * * * * * *

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